

Microfluidics

Microfluidics refers to a system that manipulates a small amount of fluids (10^{-9} to 10^{-18} liters) using small channels with sizes ten to hundreds micrometres. It is a multidisciplinary field that involves molecular analysis, molecular biology, and microelectronics.^[1] It has practical applications in the design of systems that process low volumes of fluids to achieve multiplexing, automation, and high-throughput screening. Microfluidics emerged in the beginning of the 1980s and is used in the development of inkjet printheads, DNA chips, lab-on-a-chip technology, micro-propulsion, and micro-thermal technologies.

Typically, micro means one of the following features:

- Small volumes (μL, nL, pL, fL)
- Small size
- Low energy consumption
- Microdomain effects

Typically microfluidic systems transport, mix, separate, or otherwise process fluids. Various applications rely on passive fluid control using capillary forces, in the form of capillary flow modifying elements, akin to flow resistors and flow accelerators. In some applications, external actuation means are additionally used for a directed transport of the media. Examples are rotary drives applying centrifugal forces for the fluid transport on the passive chips. **Active microfluidics** refers to the defined manipulation of the working fluid by active (micro) components such as micropumps or microvalves. Micropumps supply fluids in a continuous manner or are used for dosing. Microvalves determine the flow direction or the mode of movement of pumped liquids. Often, processes normally carried out in a lab are miniaturised on a single chip, which enhances efficiency and mobility, and reduces sample and reagent volumes.

Microscale behaviour of fluids

The behaviour of fluids at the microscale can differ from "macrofluidic" behaviour in that factors such as surface tension, energy dissipation, and fluidic resistance start to dominate the system. Microfluidics studies how these behaviours change, and how they can be worked around, or exploited for new uses.^{[2][3][4][5][6]}

At small scales (channel size of around 100 nanometers to 500 micrometers) some interesting and sometimes unintuitive properties appear. In particular, the Reynolds number (which compares the effect of the momentum of a fluid to the effect of viscosity) can become very low. A key consequence is co-flowing fluids do not necessarily mix in the traditional sense, as flow becomes laminar rather than turbulent; molecular transport between them must often be through diffusion.^[7]

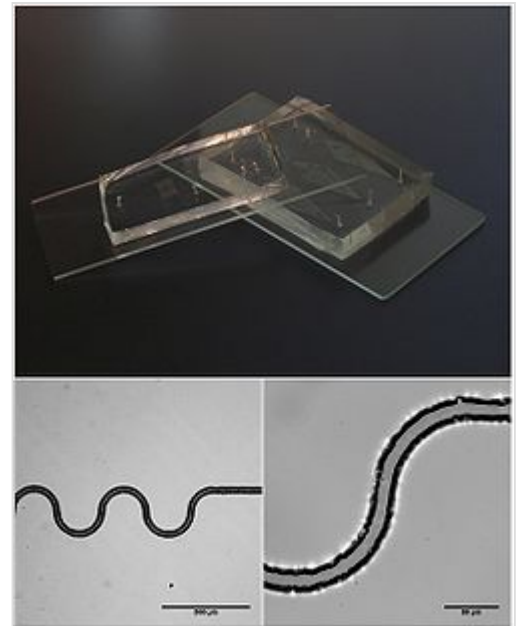
High specificity of chemical and physical properties (concentration, pH, temperature, shear force, etc.) can also be ensured resulting in more uniform reaction conditions and higher grade products in single and multi-step reactions.^{[8][9]}

Various kinds of microfluidic flows

Microfluidic flows need only be constrained by geometrical length scale – the modalities and methods used to achieve such a geometrical constraint are highly dependent on the targeted application.^[10] Traditionally, microfluidic flows have been generated inside closed channels with the channel cross section being in the order of 10 μm x 10 μm . Each of these methods has its own associated techniques to maintain robust fluid flow which have matured over several years.

Open microfluidics

The behavior of fluids and their control in open microchannels was pioneered around 2005^[11] and applied in air-to-liquid sample collection^{[12][13]} and chromatography.^[14] In open microfluidics, at least one boundary of the system is removed, exposing the fluid to air or another interface (i.e. liquid).^{[15][16][17]} Advantages of open microfluidics include accessibility to the flowing liquid for intervention, larger liquid-gas surface area, and minimized bubble formation.^{[18][15][17][19]} Another advantage of open microfluidics is the ability to integrate open systems with surface-tension driven fluid flow, which eliminates the need for external pumping methods such as peristaltic or syringe pumps.^[20] Open microfluidic devices are also easy and inexpensive to fabricate by milling, thermoforming, and hot embossing.^{[21][22][23][24]} In addition, open microfluidics eliminates the need to glue or bond a cover for devices, which could be detrimental to capillary flows. Examples of open microfluidics include open-channel microfluidics, rail-based microfluidics, paper-based, and thread-based microfluidics.^{[15][20][25]} Disadvantages to open systems include susceptibility to evaporation,^[26] contamination,^[27] and limited flow rate.^[17]



Silicone rubber and glass microfluidic devices. Top: a photograph of the devices. Bottom: Phase contrast micrographs of a serpentine channel ~15 μm wide.

Continuous-flow microfluidics

Continuous flow microfluidics rely on the control of a steady state liquid flow through narrow channels or porous media predominantly by accelerating or hindering fluid flow in capillary elements.^[28] In paper based microfluidics, capillary elements can be achieved through the simple variation of section geometry. In general, the actuation of liquid flow is implemented either by external pressure sources, external mechanical pumps, integrated mechanical micropumps, or by combinations of capillary forces and electrokinetic mechanisms.^{[29][30]} Continuous-flow microfluidic operation is the mainstream approach because it is easy to implement and less sensitive to protein fouling problems. Continuous-flow devices are adequate for many well-defined and simple biochemical applications, and for certain tasks such as chemical separation, but they are less suitable for tasks requiring a high degree of flexibility or fluid manipulations. These closed-channel systems are inherently difficult to integrate and scale because the parameters that govern flow field vary along the flow path making the fluid flow at any one location dependent on the properties of the entire system. Permanently etched microstructures also lead to limited reconfigurability and poor fault tolerance capability. Computer-aided design automation approaches for continuous-flow microfluidics have been proposed in recent years to alleviate the design effort and to solve the scalability problems.^[31]

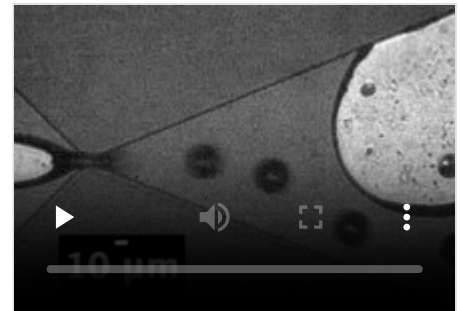
Process monitoring capabilities in continuous-flow systems can be achieved with highly sensitive microfluidic flow sensors based on MEMS technology, which offers resolutions down to the nanoliter range.

Droplet-based microfluidics

Droplet-based microfluidics is a subcategory of microfluidics in contrast with continuous microfluidics; droplet-based microfluidics manipulates discrete volumes of fluids in immiscible phases with low Reynolds number and laminar flow regimes. Interest in droplet-based microfluidics systems has been growing substantially in past decades. Microdroplets allow for handling miniature volumes (μl to fl) of fluids conveniently, provide better mixing, encapsulation, sorting, and sensing, and suit high throughput experiments.^[33] Exploiting the benefits of droplet-based microfluidics efficiently requires a deep understanding of droplet generation^[34] to perform various logical operations^{[35][36]} such as droplet manipulation,^[37] droplet sorting,^[38] droplet merging,^[39] and droplet breakup.^[40]



micro fluid sensor



High frame rate video showing microbubble pinch-off formation in a flow-focusing microfluidic device^[32]

Digital microfluidics

Alternatives to the above closed-channel continuous-flow systems include novel open structures, where discrete, independently controllable droplets are manipulated on a substrate using electrowetting. Following the analogy of digital microelectronics, this approach is referred to as digital microfluidics. Le Pesant et al. pioneered the use of electrocapillary forces to move droplets on a digital track.^[41] The "fluid transistor" pioneered by Cytonix^[42] also played a role. The technology was subsequently commercialised by Duke University. By using discrete unit-volume droplets,^[34] a microfluidic function can be reduced to a set of repeated basic operations, i.e., moving one unit of fluid over one unit of distance. This "digitisation" method facilitates the use of a hierarchical and cell-based approach for microfluidic biochip design. Therefore, digital microfluidics offers a flexible and scalable system architecture as well as high fault-tolerance capability. Moreover, because each droplet can be controlled independently, these systems also have dynamic reconfigurability, whereby groups of unit cells in a microfluidic array can be reconfigured to change their functionality during the concurrent execution of a set of bioassays. Although droplets are manipulated in confined microfluidic channels, since the control on droplets is not independent, it should not be confused as "digital microfluidics". One common actuation method for digital microfluidics is electrowetting-on-dielectric (EWOD).^[43] Many lab-on-a-chip applications have been demonstrated within the digital microfluidics paradigm using electrowetting. However, recently other techniques for droplet manipulation have also been demonstrated using magnetic force,^[44] surface acoustic waves,^[45] optoelectrowetting, mechanical actuation,^[46] etc.

Paper-based microfluidics

Paper-based microfluidic devices fill a growing niche for portable, cheap, and user-friendly medical diagnostic systems.^[47] Paper based microfluidics rely on the phenomenon of capillary penetration in porous media.^[48] To tune fluid penetration in porous substrates such as paper in two and three dimensions, the pore structure, wettability and geometry of the microfluidic devices can be controlled while the viscosity and evaporation rate of the liquid play a further significant role. Many such devices feature hydrophobic barriers on hydrophilic paper that passively transport aqueous solutions to outlets where biological reactions

take place.^[49] Paper-based microfluidics are considered as portable point-of-care biosensors used in a remote setting where advanced medical diagnostic tools are not accessible.^[50] Current applications include portable glucose detection^[51] and environmental testing,^[52] with hopes of reaching areas that lack advanced medical diagnostic tools.

Particle detection microfluidics

One application area that has seen significant academic effort and some commercial effort is in the area of particle detection in fluids. Particle detection of small fluid-borne particles down to about 1 μm in diameter is typically done using a Coulter counter, in which electrical signals are generated when a weakly-conducting fluid such as in saline water is passed through a small ($\sim 100\text{ }\mu\text{m}$ diameter) pore, so that an electrical signal is generated that is directly proportional to the ratio of the particle volume to the pore volume. The physics behind this is relatively simple, described in a classic paper by DeBlois and Bean,^[53] and the implementation first described in Coulter's original patent.^[54] This is the method used to e.g. size and count erythrocytes (red blood cells [wiki]) as well as leukocytes (white blood cells) for standard blood analysis. The generic term for this method is resistive pulse sensing (RPS); Coulter counting is a trademark term. However, the RPS method does not work well for particles below 1 μm diameter, as the signal-to-noise ratio falls below the reliably detectable limit, set mostly by the size of the pore in which the analyte passes and the input noise of the first-stage amplifier.

The limit on the pore size in traditional RPS Coulter counters is set by the method used to make the pores, which while a trade secret, most likely uses traditional mechanical methods. This is where microfluidics can have an impact: The lithography-based production of microfluidic devices, or more likely the production of reusable molds for making microfluidic devices using a molding process, is limited to sizes much smaller than traditional machining. Critical dimensions down to 1 μm are easily fabricated, and with a bit more effort and expense, feature sizes below 100 nm can be patterned reliably as well. This enables the inexpensive production of pores integrated in a microfluidic circuit where the pore diameters can reach sizes of order 100 nm, with a concomitant reduction in the minimum particle diameters by several orders of magnitude.

As a result there has been some university-based development of microfluidic particle counting and sizing^{[55][56][57][58][59][60][61][62][63][64]} with the accompanying commercialization of this technology. This method has been termed microfluidic resistive pulse sensing (MRPS).

Microfluidic-assisted magnetophoresis

One major area of application for microfluidic devices is the separation and sorting of different fluids or cell types. Recent developments in the microfluidics field have seen the integration of microfluidic devices with magnetophoresis; the migration of particles by a magnetic field.^[65] This can be accomplished by sending a fluid containing at least one magnetic component through a microfluidic channel that has a magnet positioned along the length of the channel. This creates a magnetic field inside the microfluidic channel which draws magnetically active substances towards it, effectively separating the magnetic and non-magnetic components of the fluid. This technique can be readily utilized in industrial settings where the fluid at hand already contains magnetically active material. For example, a handful of metallic impurities can find their way into certain consumable liquids, namely milk and other dairy products.^[66] Conveniently, in the case of milk, many of these metal contaminants exhibit paramagnetism. Therefore, before packaging, milk can be flowed through channels with magnetic gradients as a means of purifying out the metal contaminants.

Other, more research-oriented applications of microfluidic-assisted magnetophoresis are numerous and are generally targeted towards cell separation. The general way this is accomplished involves several steps. First, a paramagnetic substance (usually micro/nanoparticles or a paramagnetic fluid)^[67] needs to be functionalized to target the cell type of interest. This can be accomplished by identifying a transmembrane protein unique to the cell type of interest and subsequently functionalizing magnetic particles with the complementary antigen or antibody.^{[66][68][69][70][71]} Once the magnetic particles are functionalized, they are dispersed in a cell mixture where they bind to only the cells of interest. The resulting cell/particle mixture can then be flowed through a microfluidic device with a magnetic field to separate the targeted cells from the rest.

Conversely, microfluidic-assisted magnetophoresis may be used to facilitate efficient mixing within microdroplets or plugs. To accomplish this, microdroplets are injected with paramagnetic nanoparticles and are flowed through a straight channel which passes through rapidly alternating magnetic fields. This causes the magnetic particles to be quickly pushed from side to side within the droplet and results in the mixing of the microdroplet contents.^[70] This eliminates the need for tedious engineering considerations that are necessary for traditional, channel-based droplet mixing. Other research has also shown that the label-free separation of cells may be possible by suspending cells in a paramagnetic fluid and taking advantage of the magneto-Archimedes effect.^{[72][73]} While this does eliminate the complexity of particle functionalization, more research is needed to fully understand the magneto-Archimedes phenomenon and how it can be used to this end. This is not an exhaustive list of the various applications of microfluidic-assisted magnetophoresis; the above examples merely highlight the versatility of this separation technique in both current and future applications.

Key application areas

Microfluidic structures include micropneumatic systems, i.e. microsystems for the handling of off-chip fluids (liquid pumps, gas valves, etc.), and microfluidic structures for the on-chip handling of nanoliter (nl) and picoliter (pl) volumes.^[74] To date, the most successful commercial application of microfluidics is the inkjet printhead.^[75] Additionally, microfluidic manufacturing advances mean that makers can produce the devices in low-cost plastics^[76] and automatically verify part quality.^[77]

Advances in microfluidics technology are revolutionizing molecular biology procedures for enzymatic analysis (e.g., glucose and lactate assays), DNA analysis (e.g., polymerase chain reaction and high-throughput sequencing), proteomics, and in chemical synthesis.^{[28][78]} The basic idea of microfluidic biochips is to integrate assay operations such as detection, as well as sample pre-treatment and sample preparation on one chip.^{[79][80]}

An emerging application area for biochips is clinical pathology, especially the immediate point-of-care diagnosis of diseases.^[81] In addition, microfluidics-based devices, capable of continuous sampling and real-time testing of air/water samples for biochemical toxins and other dangerous pathogens,^[82] can serve as an always-on "bio-smoke alarm" for early warning.

Microfluidic technology has led to the creation of powerful tools for biologists to control the complete cellular environment, leading to new questions and discoveries. Many diverse advantages of this technology for microbiology are listed below:

- General single cell studies including growth^{[83][33]}
- Cellular aging: microfluidic devices such as the "mother machine" allow tracking of thousands of individual cells for many generations until they die^[83]

- Microenvironmental control: ranging from mechanical environment^[84] to chemical environment^{[85][86]}
- Precise spatiotemporal concentration gradients by incorporating multiple chemical inputs to a single device^[87]
- Force measurements of adherent cells or confined chromosomes: objects trapped in a microfluidic device can be directly manipulated using optical tweezers or other force-generating methods^[88]
- Confining cells and exerting controlled forces by coupling with external force-generation methods such as Stokes flow, optical tweezer, or controlled deformation of the PDMS (Polydimethylsiloxane) device^{[88][89][90]}
- Electric field integration^[90]
- Plant on a chip and plant tissue culture^[91]
- Antibiotic resistance: microfluidic devices can be used as heterogeneous environments for microorganisms. In a heterogeneous environment, it is easier for a microorganism to evolve. This can be useful for testing the acceleration of evolution of a microorganism / for testing the development of antibiotic resistance.

Some of these areas are further elaborated in the sections below:

DNA chips (microarrays)

Early biochips were based on the idea of a DNA microarray, e.g., the GeneChip DNAarray from Affymetrix, which is a piece of glass, plastic or silicon substrate, on which pieces of DNA (probes) are affixed in a microscopic array. Similar to a DNA microarray, a protein array is a miniature array where a multitude of different capture agents, most frequently monoclonal antibodies, are deposited on a chip surface; they are used to determine the presence and/or amount of proteins in biological samples, e.g., blood. A drawback of DNA and protein arrays is that they are neither reconfigurable nor scalable after manufacture. Digital microfluidics has been described as a means for carrying out Digital PCR.

Molecular biology

In addition to microarrays, biochips have been designed for two-dimensional electrophoresis,^[92] transcriptome analysis,^[93] and PCR amplification.^[94] Other applications include various electrophoresis and liquid chromatography applications for proteins and DNA, cell separation, in particular, blood cell separation, protein analysis, cell manipulation and analysis including cell viability analysis^[33] and microorganism capturing.^[80]

Evolutionary biology

By combining microfluidics with landscape ecology and nanofluidics, a nano/micro fabricated fluidic landscape can be constructed by building local patches of bacterial habitat and connecting them by dispersal corridors. The resulting landscapes can be used as physical implementations of an adaptive landscape,^[95] by generating a spatial mosaic of patches of opportunity distributed in space and time. The patchy nature of

these fluidic landscapes allows for the study of adapting bacterial cells in a metapopulation system. The evolutionary ecology of these bacterial systems in these synthetic ecosystems allows for using biophysics to address questions in evolutionary biology.

Cell behavior

The ability to create precise and carefully controlled chemoattractant gradients makes microfluidics the ideal tool to study motility,^[96] chemotaxis and the ability to evolve / develop resistance to antibiotics in small populations of microorganisms and in a short period of time. These microorganisms including bacteria^[97] and the broad range of organisms that form the marine microbial loop,^[98] responsible for regulating much of the oceans' biogeochemistry.

Microfluidics has also greatly aided the study of durotaxis by facilitating the creation of durotactic (stiffness) gradients.

Cellular biophysics

By rectifying the motion of individual swimming bacteria,^[99] microfluidic structures can be used to extract mechanical motion from a population of motile bacterial cells.^[100] This way, bacteria-powered rotors can be built.^{[101][102]}

Optics

The merger of microfluidics and optics is typically known as optofluidics. Examples of optofluidic devices are tunable microlens arrays^{[103][104]} and optofluidic microscopes.

Microfluidic flow enables fast sample throughput, automated imaging of large sample populations, as well as 3D capabilities.^{[105][106]} or superresolution.^[107]

Photonics Lab on a Chip (PhLOC)

Due to the increase in safety concerns and operating costs of common analytic methods (ICP-MS, ICP-AAS, and ICP-OES^[108]), the Photonics Lab on a Chip (PhLOC) is becoming an increasingly popular tool for the analysis of actinides and nitrates in spent nuclear waste. The PhLOC is based on the simultaneous application of Raman and UV-Vis-NIR spectroscopy,^[109] which allows for the analysis of more complex mixtures which contain several actinides at different oxidation states.^[110] Measurements made with these methods have been validated at the bulk level for industrial tests,^{[108][111]} and are observed to have a much lower variance at the micro-scale.^[112] This approach has been found to have molar extinction coefficients (UV-Vis) in line with known literature values over a comparatively large concentration span for 150 μL ^[110] via elongation of the measurement channel, and obeys Beer's Law at the micro-scale for U(IV).^[113] Through the development of a spectrophotometric approach to analyzing spent fuel, an on-line method for measurement of reactant quantities is created, increasing the rate at which samples can be analyzed and thus decreasing the size of deviations detectable within reprocessing.^[111]

Through the application of the PhLOC, flexibility and safety of operational methods are increased. Since the analysis of spent nuclear fuel involves extremely harsh conditions, the application of disposable and rapidly produced devices (Based on castable and/or engravable materials such as PDMS, PMMA, and glass^[114]) is advantageous, although material integrity must be considered under specific harsh

conditions.^[113] Through the usage of fiber optic coupling, the device can be isolated from instrumentation, preventing irradiative damage and minimizing the exposure of lab personnel to potentially harmful radiation, something not possible on the lab scale nor with the previous standard of analysis.^[110] The shrinkage of the device also allows for lower amounts of analyte to be used, decreasing the amount of waste generated and exposure to hazardous materials.^[110]

Expansion of the PhLOC to miniaturize research of the full nuclear fuel cycle is currently being evaluated, with steps of the PUREX process successfully being demonstrated at the micro-scale.^[109] Likewise, the microfluidic technology developed for the analysis of spent nuclear fuel is predicted to expand horizontally to analysis of other actinide, lanthanides, and transition metals with little to no modification.^[110]

High Performance Liquid Chromatography (HPLC)

HPLC in the field of microfluidics comes in two different forms. Early designs included running liquid through the HPLC column then transferring the eluted liquid to microfluidic chips and attaching HPLC columns to the microfluidic chip directly.^[115] The early methods had the advantage of easier detection from certain machines like those that measure fluorescence.^[116] More recent designs have fully integrated HPLC columns into microfluidic chips. The main advantage of integrating HPLC columns into microfluidic devices is the smaller form factor that can be achieved, which allows for additional features to be combined within one microfluidic chip. Integrated chips can also be fabricated from multiple different materials, including glass and polyimide which are quite different from the standard material of PDMS used in many different droplet-based microfluidic devices.^{[117][118]} This is an important feature because different applications of HPLC microfluidic chips may call for different pressures. PDMS fails in comparison for high-pressure uses compared to glass and polyimide. High versatility of HPLC integration ensures robustness by avoiding connections and fittings between the column and chip.^[119] The ability to build off said designs in the future allows the field of microfluidics to continue expanding its potential applications.

The potential applications surrounding integrated HPLC columns within microfluidic devices have proven expansive over the last 10–15 years. The integration of such columns allows for experiments to be run where materials were in low availability or very expensive, like in biological analysis of proteins. This reduction in reagent volumes allows for new experiments like single-cell protein analysis, which due to size limitations of prior devices, previously came with great difficulty.^[120] The coupling of HPLC-chip devices with other spectrometry methods like mass-spectrometry allow for enhanced confidence in identification of desired species, like proteins.^[121] Microfluidic chips have also been created with internal delay-lines that allow for gradient generation to further improve HPLC, which can reduce the need for further separations.^[122] Some other practical applications of integrated HPLC chips include the determination of drug presence in a person through their hair^[123] and the labeling of peptides through reverse phase liquid chromatography.^[124]

Acoustic droplet ejection (ADE)

Acoustic droplet ejection uses a pulse of ultrasound to move low volumes of fluids (typically nanoliters or picoliters) without any physical contact. This technology focuses acoustic energy into a fluid sample to eject droplets as small as a millionth of a millionth of a litre (picoliter = 10^{-12} litre). ADE technology is a very

gentle process, and it can be used to transfer proteins, high molecular weight DNA and live cells without damage or loss of viability. This feature makes the technology suitable for a wide variety of applications including proteomics and cell-based assays.

Fuel cells

Microfluidic fuel cells can use laminar flow to separate the fuel and its oxidant to control the interaction of the two fluids without the physical barrier that conventional fuel cells require.^{[125][126][127]}

Astrobiology

To understand the prospects for life to exist elsewhere in the universe, astrobiologists are interested in measuring the chemical composition of extraplanetary bodies.^[128] Because of their small size and wide-ranging functionality, microfluidic devices are uniquely suited for these remote sample analyses.^{[129][130][131]} From an extraterrestrial sample, the organic content can be assessed using microchip capillary electrophoresis and selective fluorescent dyes.^[132] These devices are capable of detecting amino acids,^[133] peptides,^[134] fatty acids,^[135] and simple aldehydes, ketones,^[136] and thiols.^[137] These analyses coupled together could allow powerful detection of the key components of life, and hopefully inform our search for functioning extraterrestrial life.^[138]

Food science

Microfluidic techniques such as droplet microfluidics, paper microfluidics, and lab-on-a-chip are used in the realm of food science in a variety of categories.^[139] Research in nutrition,^{[140][141]} food processing, and food safety benefit from microfluidic technique because experiments can be done with less reagents.^[139]

Food processing requires the ability to enable shelf stability in foods, such as emulsions or additions of preservatives. Techniques such as droplet microfluidics are used to create emulsions that are more controlled and complex than those created by traditional homogenization due to the precision of droplets that is achievable. Using microfluidics for emulsions is also more energy efficient compared to homogenization in which “only 5% of the supplied energy is used to generate the emulsion, with the rest dissipated as heat”^[142] Although these methods have benefits, they currently lack the ability to be produced at large scale that is needed for commercialization.^[143] Microfluidics are also used in research as they allow for innovation in food chemistry and food processing.^{[139][143]} An example in food engineering research is a novel micro-3D-printed device fabricated to research production of droplets for potential food processing industry use, particularly in work with enhancing emulsions.^[144]

Paper and droplet microfluidics allow for devices that can detect small amounts of unwanted bacteria or chemicals, making them useful in food safety and analysis.^[145] Paper-based microfluidic devices are often referred to as microfluidic paper-based analytical devices (μ PADs) and can detect such things as nitrate,^[146] preservatives,^[147] or antibiotics^[148] in meat by a colorimetric reaction that can be detected with a smartphone. These methods are being researched because they use less reactants, space, and time compared to traditional techniques such as liquid chromatography. μ PADs also make home detection tests possible, which is of interest to those with allergies and intolerances.^[146] In addition to paper-based

methods, research demonstrates droplet-based microfluidics shows promise in drastically shortening the time necessary to confirm viable bacterial contamination in agricultural waters in the domestic and international food industry.^[145]

Future directions

Microfluidics for personalized cancer treatment

Personalized cancer treatment is a tuned method based on the patient's diagnosis and background. Microfluidic technology offers sensitive detection with higher throughput, as well as reduced time and costs. For personalized cancer treatment, tumor composition and drug sensitivities are very important.^[149]

A patient's drug response can be predicted based on the status of biomarkers, or the severity and progression of the disease can be predicted based on the atypical presence of specific cells.^[150] Drop-qPCR is a droplet microfluidic technology in which droplets are transported in a reusable capillary and alternately flow through two areas maintained at different constant temperatures and fluorescence detection. It can be efficient with a low contamination risk to detect Her2.^[149] A digital droplet-based PCR method can be used to detect the KRAS mutations with TaqMan probes, to enhance detection of the mutative gene ratio.^[151] In addition, accurate prediction of postoperative disease progression in breast or prostate cancer patients is essential for determining post-surgery treatment. A simple microfluidic chamber, coated with a carefully formulated extracellular matrix mixture is used for cells obtained from tumor biopsy after 72 hours of growth and a thorough evaluation of cells by imaging.^[152]

Microfluidics is also suitable for circulating tumor cells (CTCs) and non-CTCs liquid biopsy analysis. Beads conjugate to anti-epithelial cell adhesion molecule (EpCAM) antibodies for positive selection in the CTCs isolation chip (iCHIP).^[153] CTCs can also be detected by using the acidification of the tumor microenvironment and the difference in membrane capacitance.^{[154][155]} CTCs are isolated from blood by a microfluidic device, and are cultured on-chip, which can be a method to capture more biological information in a single analysis. For example, it can be used to test the cell survival rate of 40 different drugs or drug combinations.^[156] Tumor-derived extracellular vesicles can be isolated from urine and detected by an integrated double-filtration microfluidic device; they also can be isolated from blood and detected by electrochemical sensing method with a two-level amplification enzymatic assay.^{[157][158]}

Tumor materials can directly be used for detection through microfluidic devices. To screen primary cells for drugs, it is often necessary to distinguish cancerous cells from non-cancerous cells. A microfluidic chip based on the capacity of cells to pass small constrictions can sort the cell types, metastases.^[159] Droplet-based microfluidic devices have the potential to screen different drugs or combinations of drugs, directly on the primary tumor sample with high accuracy. To improve this strategy, the microfluidic program with a sequential manner of drug cocktails, coupled with fluorescent barcodes, is more efficient.^[160] Another advanced strategy is detecting growth rates of single-cell by using suspended microchannel resonators, which can predict drug sensitivities of rare CTCs.^[161]

Microfluidics devices also can simulate the tumor microenvironment, to help to test anticancer drugs. Microfluidic devices with 2D or 3D cell cultures can be used to analyze spheroids for different cancer systems (such as lung cancer and ovarian cancer), and are essential for multiple anti-cancer drugs and toxicity tests. This strategy can be improved by increasing the throughput and production of spheroids. For example, one droplet-based microfluidic device for 3D cell culture produces 500 spheroids per chip.^[162] These spheroids can be cultured longer in different surroundings to analyze and monitor. The other

advanced technology is organs-on-a-chip, and it can be used to simulate several organs to determine the drug metabolism and activity based on vessels mimicking, as well as mimic pH, oxygen... to analyze the relationship between drugs and human organ surroundings.^[162]

A recent strategy is single-cell chromatin immunoprecipitation (ChIP)-Sequencing in droplets, which operates by combining droplet-based single cell RNA sequencing with DNA-barcoded antibodies, possibly to explore the tumor heterogeneity by the genotype and phenotype to select the personalized anti-cancer drugs and prevent the cancer relapse.^[163]

See also



Biology portal



Technology portal

- Advanced Simulation Library
- Droplet-based microfluidics
- Fluidics
- Induced-charge electrokinetics
- Lab-on-a-chip
- Microfluidic cell culture
- Microfluidic modulation spectroscopy
- Microphysiometry
- Micropumps
- Microvalves
- uFluids@Home
- Paper-based microfluidics

References

1. Whitesides, George M. (July 2006). "The origins and the future of microfluidics" (<http://www.nature.com/articles/nature05058>). *Nature*. **442** (7101): 368–373. Bibcode:2006Natur.442..368W (<https://ui.adsabs.harvard.edu/abs/2006Natur.442..368W>). doi:10.1038/nature05058 (<https://doi.org/10.1038%2Fnature05058>). ISSN 0028-0836 (<http://www.worldcat.org/issn/0028-0836>). PMID 16871203 (<https://pubmed.ncbi.nlm.nih.gov/16871203>). S2CID 205210989 (<https://api.semanticscholar.org/CorpusID:205210989>).
2. Terry SC, Jerman JH, Angell JB (December 1979). "A gas chromatographic air analyzer fabricated on a silicon wafer". *IEEE Transactions on Electron Devices*. **26** (12): 1880–6. Bibcode:1979ITED...26.1880T (<https://ui.adsabs.harvard.edu/abs/1979ITED...26.1880T>). doi:10.1109/T-ED.1979.19791 (<https://doi.org/10.1109%2FT-ED.1979.19791>). S2CID 21971431 (<https://api.semanticscholar.org/CorpusID:21971431>).
3. Kirby BJ (2010). *Micro- and Nanoscale Fluid Mechanics: Transport in Microfluidic Devices* (<https://web.archive.org/web/20190428234717/http://www.kirbyresearch.com/textbook/>). Cambridge University Press. Archived from the original (<http://www.kirbyresearch.com/textbook>) on 2019-04-28. Retrieved 2010-02-13.
4. Karniadakis GM, Beskok A, Aluru N (2005). *Microflows and Nanoflows*. Springer Verlag.
5. Bruus H (2007). *Theoretical Microfluidics*. Oxford University Press.
6. Shkolnikov V (2019). *Principles of Microfluidics*. ISBN 978-1790217281.

7. Tabeling P (2005). *Introduction to Microfluidics* (<https://archive.org/details/introductiontomi000tabe>). Oxford University Press. ISBN 978-0-19-856864-3.
8. Chokkalingam V, Weidenhof B, Krämer M, Maier WF, Herminghaus S, Seemann R (July 2010). "Optimized droplet-based microfluidics scheme for sol-gel reactions". *Lab on a Chip*. **10** (13): 1700–1705. doi:10.1039/b926976b (<https://doi.org/10.1039%2Fb926976b>). PMID 20405061 (<https://pubmed.ncbi.nlm.nih.gov/20405061>).
9. Shestopalov I, Tice JD, Ismagilov RF (August 2004). "Multi-step synthesis of nanoparticles performed on millisecond time scale in a microfluidic droplet-based system" (<https://authors.library.caltech.edu/40869/>). *Lab on a Chip*. **4** (4): 316–321. doi:10.1039/b403378g (<https://doi.org/10.1039%2Fb403378g>). PMID 15269797 (<https://pubmed.ncbi.nlm.nih.gov/15269797>).
10. Thomas DJ, McCall C, Tehrani Z, Claypole TC (June 2017). "Three-Dimensional–Printed Laboratory-on-a-Chip With Microelectronics and Silicon Integration" (<https://cronfa.swan.ac.uk/Record/cronfa34529>). *Point of Care*. **16** (2): 97–101. doi:10.1097/POC.0000000000000132 (<https://doi.org/10.1097%2FPOC.0000000000000132>). S2CID 58306257 (<https://api.semanticscholar.org/CorpusID:58306257>).
11. Melin J, van der Wijngaart W, Stemme G (June 2005). "Behaviour and design considerations for continuous flow closed-open-closed liquid microchannels" (<http://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-14775>). *Lab on a Chip*. **5** (6): 682–686. doi:10.1039/b501781e (<https://doi.org/10.1039%2Fb501781e>). PMID 15915262 (<https://pubmed.ncbi.nlm.nih.gov/15915262>).
12. Frisk T, Rönnholm D, van der Wijngaart W, Stemme G (December 2006). "A micromachined interface for airborne sample-to-liquid transfer and its application in a biosensor system" (<http://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-14188>). *Lab on a Chip*. **6** (12): 1504–1509. doi:10.1039/B612526N (<https://doi.org/10.1039%2FB612526N>). PMID 17203153 (<https://pubmed.ncbi.nlm.nih.gov/17203153>).
13. Frisk T, Sandström N, Eng L, van der Wijngaart W, Månsson P, Stemme G (October 2008). "An integrated QCM-based narcotics sensing microsystem" (<http://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-14189>). *Lab on a Chip*. **8** (10): 1648–1657. doi:10.1039/b800487k (<https://doi.org/10.1039%2Fb800487k>). PMID 18813386 (<https://pubmed.ncbi.nlm.nih.gov/18813386>).
14. Jacksén J, Frisk T, Redeby T, Parmar V, van der Wijngaart W, Stemme G, Emmer A (July 2007). "Off-line integration of CE and MALDI-MS using a closed-open-closed microchannel system" (<https://doi.org/10.1002%2Felps.200600735>). *Electrophoresis*. **28** (14): 2458–2465. doi:10.1002/elps.200600735 (<https://doi.org/10.1002%2Felps.200600735>). PMID 17577881 (<https://pubmed.ncbi.nlm.nih.gov/17577881>). S2CID 16337938 (<https://api.semanticscholar.org/CorpusID:16337938>).
15. Berthier J, Brakke KA, Berthier E (2016-08-01). *Open Microfluidics*. doi:10.1002/9781118720936 (<https://doi.org/10.1002%2F9781118720936>). ISBN 9781118720936.
16. Pfohl T, Mugele F, Seemann R, Herminghaus S (December 2003). "Trends in microfluidics with complex fluids" (https://ris.utwente.nl/ws/files/6488126/trends_in_microfluidics.pdf) (PDF). *ChemPhysChem*. **4** (12): 1291–1298. doi:10.1002/cphc.200300847 (<https://doi.org/10.1002%2Fcphc.200300847>). PMID 14714376 (<https://pubmed.ncbi.nlm.nih.gov/14714376>).
17. Kaigala GV, Lovchik RD, Delamarche E (November 2012). "Microfluidics in the "open space" for performing localized chemistry on biological interfaces". *Angewandte Chemie*. **51** (45): 11224–11240. doi:10.1002/anie.201201798 (<https://doi.org/10.1002%2Fanie.201201798>). PMID 23111955 (<https://pubmed.ncbi.nlm.nih.gov/23111955>).
18. Lade, R. K.; Jochem, K. S.; Macosko, C. W.; Francis, L. F. (2018). "Capillary Coatings: Flow and Drying Dynamics in Open Microchannels" (<https://doi.org/10.1021/acs.langmuir.8b00811>). *Langmuir*. **34** (26): 7624–7639. doi:10.1021/acs.langmuir.8b00811 (<https://doi.org/10.1021%2Facs.langmuir.8b00811>). PMID 29787270 (<https://pubmed.ncbi.nlm.nih.gov/29787270>).

19. Li C, Boban M, Tuteja A (April 2017). "Open-channel, water-in-oil emulsification in paper-based microfluidic devices". *Lab on a Chip*. **17** (8): 1436–1441. doi:10.1039/c7lc00114b (<https://doi.org/10.1039%2Fc7lc00114b>). PMID 28322402 (<https://pubmed.ncbi.nlm.nih.gov/28322402>). S2CID 5046916 (<https://api.semanticscholar.org/CorpusID:5046916>).
20. Casavant BP, Berthier E, Theberge AB, Berthier J, Montanez-Sauri SI, Bischel LL, et al. (June 2013). "Suspended microfluidics" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3690848>). *Proceedings of the National Academy of Sciences of the United States of America*. **110** (25): 10111–10116. Bibcode:2013PNAS..11010111C (<https://ui.adsabs.harvard.edu/abs/2013PNAS..11010111C>). doi:10.1073/pnas.1302566110 (<https://doi.org/10.1073%2Fpnas.1302566110>). PMC 3690848 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3690848>). PMID 23729815 (<https://pubmed.ncbi.nlm.nih.gov/23729815>).
21. Guckenberger DJ, de Groot TE, Wan AM, Beebe DJ, Young EW (June 2015). "Micromilling: a method for ultra-rapid prototyping of plastic microfluidic devices" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4439323>). *Lab on a Chip*. **15** (11): 2364–2378. doi:10.1039/c5lc00234f (<https://doi.org/10.1039%2Fc5lc00234f>). PMC 4439323 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4439323>). PMID 25906246 (<https://pubmed.ncbi.nlm.nih.gov/25906246>).
22. Truckenmüller R, Rummler Z, Schaller T, Schomburg WK (2002-06-13). "Low-cost thermoforming of micro fluidic analysis chips". *Journal of Micromechanics and Microengineering*. **12** (4): 375–379. Bibcode:2002JMiMi..12..375T (<https://ui.adsabs.harvard.edu/abs/2002JMiMi..12..375T>). doi:10.1088/0960-1317/12/4/304 (<https://doi.org/10.1088%2F0960-1317%2F12%2F4%2F304>). ISSN 0960-1317 (<https://www.worldcat.org/issn/0960-1317>). S2CID 250860338 (<https://api.semanticscholar.org/CorpusID:250860338>).
23. Jeon JS, Chung S, Kamm RD, Charest JL (April 2011). "Hot embossing for fabrication of a microfluidic 3D cell culture platform" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3117225>). *Biomedical Microdevices*. **13** (2): 325–333. doi:10.1007/s10544-010-9496-0 (<https://doi.org/10.1007%2Fs10544-010-9496-0>). PMC 3117225 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3117225>). PMID 21113663 (<https://pubmed.ncbi.nlm.nih.gov/21113663>).
24. Young EW, Berthier E, Guckenberger DJ, Sackmann E, Lamers C, Meyvantsson I, et al. (February 2011). "Rapid prototyping of arrayed microfluidic systems in polystyrene for cell-based assays" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3052265>). *Analytical Chemistry*. **83** (4): 1408–1417. doi:10.1021/ac102897h (<https://doi.org/10.1021%2Fac102897h>). PMC 3052265 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3052265>). PMID 21261280 (<https://pubmed.ncbi.nlm.nih.gov/21261280>).
25. Bouaidat S, Hansen O, Bruus H, Berendsen C, Bau-Madsen NK, Thomsen P, et al. (August 2005). "Surface-directed capillary system; theory, experiments and applications". *Lab on a Chip*. **5** (8): 827–836. doi:10.1039/b502207j (<https://doi.org/10.1039%2Fb502207j>). PMID 16027933 (<https://pubmed.ncbi.nlm.nih.gov/16027933>). S2CID 18125405 (<https://api.semanticscholar.org/CorpusID:18125405>).
26. Kachel S, Zhou Y, Scharfer P, Vrančić C, Petrich W, Schabel W (February 2014). "Evaporation from open microchannel grooves". *Lab on a Chip*. **14** (4): 771–778. doi:10.1039/c3lc50892g (<https://doi.org/10.1039%2Fc3lc50892g>). PMID 24345870 (<https://pubmed.ncbi.nlm.nih.gov/24345870>).
27. Ogawa M, Higashi K, Miki N (August 2015). "Development of hydrogel microtubes for microbe culture in open environment". *2015 37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. Vol. 2015. pp. 5896–5899. doi:10.1109/EMBC.2015.7319733 (<https://doi.org/10.1109%2FEMBC.2015.7319733>). ISBN 978-1-4244-9271-8. PMID 26737633 (<https://pubmed.ncbi.nlm.nih.gov/26737633>). S2CID 4089852 (<https://api.semanticscholar.org/CorpusID:4089852>).

28. Konda A, Morin SA (June 2017). "Flow-directed synthesis of spatially variant arrays of branched zinc oxide mesostructures". *Nanoscale*. **9** (24): 8393–8400. doi:10.1039/C7NR02655B (<https://doi.org/10.1039%2FC7NR02655B>). PMID 28604901 (<http://pubmed.ncbi.nlm.nih.gov/28604901>).
29. Chang HC, Yeo L (2009). *Electrokinetically Driven Microfluidics and Nanofluidics*. Cambridge University Press.
30. "fluid transistor" (<https://web.archive.org/web/20110708215908/http://www.cytonix.com/fluid%20transistor.html>). Archived from the original (<http://www.cytonix.com/fluid%20transistor.html>) on July 8, 2011.
31. Tseng TM, Li M, Freitas DN, McAuley T, Li B, Ho TY, Araci IE, Schlichtmann U (2018). "Columba 2.0: A Co-Layout Synthesis Tool for Continuous-Flow Microfluidic Biochips". *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*. **37** (8): 1588–1601. doi:10.1109/TCAD.2017.2760628 (<https://doi.org/10.1109%2FTCAD.2017.2760628>). S2CID 49893963 (<https://api.semanticscholar.org/CorpusID:49893963>).
32. Churchman AH (2018). "Data associated with 'Combined flow-focus and self-assembly routes for the formation of lipid stabilized oil-shelled microbubbles'" (<https://archive.researchdata.leeds.ac.uk/327/>). University of Leeds. doi:10.5518/153 (<https://doi.org/10.5518%2F153>).
33. Chokkalingam V, Tel J, Wimmers F, Liu X, Semenov S, Thiele J, et al. (December 2013). "Probing cellular heterogeneity in cytokine-secreting immune cells using droplet-based microfluidics". *Lab on a Chip*. **13** (24): 4740–4744. doi:10.1039/C3LC50945A (<https://doi.org/10.1039%2FC3LC50945A>). PMID 24185478 (<https://pubmed.ncbi.nlm.nih.gov/24185478>). S2CID 46363431 (<https://api.semanticscholar.org/CorpusID:46363431>).
34. Chokkalingam V, Herminghaus S, Seemann R (2008). "Self-synchronizing Pairwise Production of Monodisperse Droplets by Microfluidic Step Emulsification" (https://archive.today/20130113004540/http://apl.aip.org/applab/v93/i25/p254101_s1). *Applied Physics Letters*. **93** (25): 254101. Bibcode:2008ApPhL..93y4101C (<https://ui.adsabs.harvard.edu/abs/2008ApPhL..93y4101C>). doi:10.1063/1.3050461 (<https://doi.org/10.1063%2F1.3050461>). Archived from the original (http://apl.aip.org/applab/v93/i25/p254101_s1) on 2013-01-13.
35. Teh SY, Lin R, Hung LH, Lee AP (February 2008). "Droplet microfluidics". *Lab on a Chip*. **8** (2): 198–220. doi:10.1039/B715524G (<https://doi.org/10.1039%2FB715524G>). PMID 18231657 (<https://pubmed.ncbi.nlm.nih.gov/18231657>). S2CID 18158748 (<https://api.semanticscholar.org/CorpusID:18158748>).
36. Prakash M, Gershenfeld N (February 2007). "Microfluidic bubble logic". *Science*. **315** (5813): 832–835. Bibcode:2007Sci...315..832P (<https://ui.adsabs.harvard.edu/abs/2007Sci...315..832P>). CiteSeerX 10.1.1.673.2864 (<https://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.673.2864>). doi:10.1126/science.1136907 (<https://doi.org/10.1126%2Fscience.1136907>). PMID 17289994 (<https://pubmed.ncbi.nlm.nih.gov/17289994>). S2CID 5882836 (<https://api.semanticscholar.org/CorpusID:5882836>).
37. Tenje M, Fornell A, Ohlin M, Nilsson J (February 2018). "Particle Manipulation Methods in Droplet Microfluidics" (<https://doi.org/10.1021%2Facs.analchem.7b01333>). *Analytical Chemistry*. **90** (3): 1434–1443. doi:10.1021/acs.analchem.7b01333 (<https://doi.org/10.1021%2Facs.analchem.7b01333>). PMID 29188994 (<https://pubmed.ncbi.nlm.nih.gov/29188994>). S2CID 46777312 (<https://api.semanticscholar.org/CorpusID:46777312>).
38. Xi HD, Zheng H, Guo W, Gañán-Calvo AM, Ai Y, Tsao CW, et al. (February 2017). "Active droplet sorting in microfluidics: a review". *Lab on a Chip*. **17** (5): 751–771. doi:10.1039/C6LC01435F (<https://doi.org/10.1039%2FC6LC01435F>). PMID 28197601 (<http://pubmed.ncbi.nlm.nih.gov/28197601>).
39. Niu X, Gulati S, Edel JB, deMello AJ (November 2008). "Pillar-induced droplet merging in microfluidic circuits". *Lab on a Chip*. **8** (11): 1837–1841. doi:10.1039/b813325e (<https://doi.org/10.1039%2Fb813325e>). PMID 18941682 (<https://pubmed.ncbi.nlm.nih.gov/18941682>).

40. Samie M, Salari A, Shafii MB (May 2013). "Breakup of microdroplets in asymmetric T junctions". *Physical Review E*. **87** (5): 053003. Bibcode:2013PhRvE..87e3003S (<https://ui.adsabs.harvard.edu/abs/2013PhRvE..87e3003S>). doi:10.1103/PhysRevE.87.053003 (<https://doi.org/10.1103/PhysRevE.87.053003>). PMID 23767616 (<https://pubmed.ncbi.nlm.nih.gov/23767616>).
41. Le Pesant et al., Electrodes for a device operating by electrically controlled fluid displacement, U.S. Pat. No. 4,569,575 (<https://worldwide.espacenet.com/patent/search/family/009290366/publication/US4569575A?q=prn%3DUS4569575>), Feb. 11, 1986.
42. NSF Award Search: Advanced Search Results (<https://www.nsf.gov/awardsearch/piSearch.do;jsessionid=D05E82394F781CBA17DB0C5AC8E3C0B8?SearchType=piSearch&page=1&QueryText=&PIFirstName=james&PILastName=brown&PIInstitution=cytonix&PIState=MD&PIZip=&PICountry=US&RestrictExpired=on&Search=Search#results>)
43. Lee J, Kim CJ (June 2000). "Surface-tension-driven microactuation based on continuous electrowetting". *Journal of Microelectromechanical Systems*. **9** (2): 171–180. doi:10.1109/84.846697 (<https://doi.org/10.1109/84.846697>). ISSN 1057-7157 (<https://www.worldcat.org/issn/1057-7157>). S2CID 25996316 (<https://api.semanticscholar.org/CorpusID:25996316>).
44. Zhang Y, Nguyen NT (March 2017). "Magnetic digital microfluidics – a review". *Lab on a Chip*. **17** (6): 994–1008. doi:10.1039/c7lc00025a (<https://doi.org/10.1039/c7lc00025a>). hdl:10072/344389 (<https://hdl.handle.net/10072/344389>). PMID 28220916 (<https://pubmed.ncbi.nlm.nih.gov/28220916>). S2CID 5013542 (<https://api.semanticscholar.org/CorpusID:5013542>).
45. Shilton RJ, Travagliati M, Beltram F, Cecchini M (August 2014). "Nanoliter-droplet acoustic streaming via ultra high frequency surface acoustic waves" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4173126>). *Advanced Materials*. **26** (29): 4941–4946. Bibcode:2014AdM....26.4941S (<https://ui.adsabs.harvard.edu/abs/2014AdM....26.4941S>). doi:10.1002/adma.201400091 (<https://doi.org/10.1002/adma.201400091>). PMC 4173126 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4173126>). PMID 24677370 (<https://pubmed.ncbi.nlm.nih.gov/24677370>).
46. Shemesh J, Bransky A, Khoury M, Levenberg S (October 2010). "Advanced microfluidic droplet manipulation based on piezoelectric actuation". *Biomedical Microdevices*. **12** (5): 907–914. doi:10.1007/s10544-010-9445-y (<https://doi.org/10.1007/s10544-010-9445-y>). PMID 20559875 (<https://pubmed.ncbi.nlm.nih.gov/20559875>). S2CID 5298534 (<https://api.semanticscholar.org/CorpusID:5298534>).
47. Berthier J, Brakke KA, Berthier E (2016). *Open Microfluidics*. John Wiley & Sons, Inc. pp. 229–256. doi:10.1002/9781118720936.ch7 (<https://doi.org/10.1002/9781118720936.ch7>). ISBN 9781118720936.
48. Liu M, Suo S, Wu J, Gan Y, Ah Hanaor D, Chen CQ (March 2019). "Tailoring porous media for controllable capillary flow". *Journal of Colloid and Interface Science*. **539**: 379–387. arXiv:2106.03526 (<https://arxiv.org/abs/2106.03526>). Bibcode:2019JCIS..539..379L (<https://ui.adsabs.harvard.edu/abs/2019JCIS..539..379L>). doi:10.1016/j.jcis.2018.12.068 (<https://doi.org/10.1016/j.jcis.2018.12.068>). PMID 30594833 (<https://pubmed.ncbi.nlm.nih.gov/30594833>). S2CID 58553777 (<https://api.semanticscholar.org/CorpusID:58553777>).
49. Galindo-Rosales FJ (2017-05-26). *Complex Fluid-Flows in Microfluidics* (<https://books.google.com/books?id=5YQIDwAAQBAJ&q=Microfabrication+Techniques+for+Microfluidic+Devices+Silverio&pg=PA24>). Springer. ISBN 9783319595931.
50. Loo J, Ho A, Turner A, Mak WC (2019). "Integrated Printed Microfluidic Biosensors". *Trends in Biotechnology*. **37** (10): 1104–1120. doi:10.1016/j.tibtech.2019.03.009 (<https://doi.org/10.1016/j.tibtech.2019.03.009>). hdl:1826/15985 (<https://hdl.handle.net/1826/15985>). PMID 30992149 (<https://pubmed.ncbi.nlm.nih.gov/30992149>). S2CID 119536401 (<https://api.semanticscholar.org/CorpusID:119536401>).

51. Martinez AW, Phillips ST, Butte MJ, Whitesides GM (2007). "Patterned paper as a platform for inexpensive, low-volume, portable bioassays" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3804133>). *Angewandte Chemie*. **46** (8): 1318–1320. doi:10.1002/anie.200603817 (<https://doi.org/10.1002%2Fanie.200603817>). PMC 3804133 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3804133>). PMID 17211899 (<https://pubmed.ncbi.nlm.nih.gov/17211899>).
52. Park TS, Yoon JY (2015-03-01). "Smartphone Detection of Escherichia coli From Field Water Samples on Paper Microfluidics" (<https://www.researchgate.net/publication/271508549>). *IEEE Sensors Journal*. **15** (3): 1902. Bibcode:2015ISenJ..15.1902P (<https://ui.adsabs.harvard.edu/abs/2015ISenJ..15.1902P>). doi:10.1109/JSEN.2014.2367039 (<https://doi.org/10.1109%2FJSEN.2014.2367039>). S2CID 34581378 (<https://api.semanticscholar.org/CorpusID:34581378>).
53. DeBlois RW, Bean CP (1970). "Counting and sizing of submicron particles by the resistive pulse technique". *Rev. Sci. Instrum.* **41** (7): 909–916. Bibcode:1970RSci...41..909D (<https://ui.adsabs.harvard.edu/abs/1970RSci...41..909D>). doi:10.1063/1.1684724 (<https://doi.org/10.1063%2F1.1684724>).
54. US 2656508 (<https://worldwide.espacenet.com/textdoc?DB=EPODOC&IDX=US2656508>), Wallace H. Coulter, "Means for counting particles suspended in a fluid", published Oct. 20, 1953
55. Kasianowicz JJ, Brandin E, Branton D, Deamer DW (November 1996). "Characterization of individual polynucleotide molecules using a membrane channel" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC19421>). *Proceedings of the National Academy of Sciences of the United States of America*. **93** (24): 13770–13773. Bibcode:1996PNAS...9313770K (<https://ui.adsabs.harvard.edu/abs/1996PNAS...9313770K>). doi:10.1073/pnas.93.24.13770 (<https://doi.org/10.1073%2Fpnas.93.24.13770>). PMC 19421 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC19421>). PMID 8943010 (<https://pubmed.ncbi.nlm.nih.gov/8943010>).
56. Li J, Gershow M, Stein D, Brandin E, Golovchenko JA (September 2003). "DNA molecules and configurations in a solid-state nanopore microscope" (<http://nrs.harvard.edu/urn-3:HUL.InstRepos:27877632>). *Nature Materials*. **2** (9): 611–615. Bibcode:2003NatMa...2..611L (<https://ui.adsabs.harvard.edu/abs/2003NatMa...2..611L>). doi:10.1038/nmat965 (<https://doi.org/10.1038%2Fnm965>). PMID 12942073 (<https://pubmed.ncbi.nlm.nih.gov/12942073>). S2CID 7521907 (<https://api.semanticscholar.org/CorpusID:7521907>).
57. Uram JD, Ke K, Hunt AJ, Mayer M (March 2006). "Label-free affinity assays by rapid detection of immune complexes in submicrometer pores". *Angewandte Chemie*. **45** (14): 2281–2285. doi:10.1002/anie.200502862 (<https://doi.org/10.1002%2Fanie.200502862>). hdl:2027.42/50668 (<https://hdl.handle.net/2027.42%2F50668>). PMID 16506296 (<https://pubmed.ncbi.nlm.nih.gov/16506296>).
58. Saleh O, Sohn LL (2003). "An artificial nanopore for molecular sensing". *Nano Lett.* **3** (1): 37–38. Bibcode:2003NanoL...3...37S (<https://ui.adsabs.harvard.edu/abs/2003NanoL...3...37S>). doi:10.1021/nl0255202 (<https://doi.org/10.1021%2Fnl0255202>).
59. Sen YH, Karnik R (May 2009). "Investigating the translocation of lambda-DNA molecules through PDMS nanopores". *Analytical and Bioanalytical Chemistry*. **394** (2): 437–446. doi:10.1007/s00216-008-2529-3 (<https://doi.org/10.1007%2Fs00216-008-2529-3>). hdl:1721.1/51892 (<https://hdl.handle.net/1721.1%2F51892>). PMID 19050856 (<https://pubmed.ncbi.nlm.nih.gov/19050856>). S2CID 7442686 (<https://api.semanticscholar.org/CorpusID:7442686>).
60. Lewpiriyawong N, Kandaswamy K, Yang C, Ivanov V, Stocker R (December 2011). "Microfluidic characterization and continuous separation of cells and particles using conducting poly(dimethyl siloxane) electrode induced alternating current-dielectrophoresis". *Analytical Chemistry*. **83** (24): 9579–9585. doi:10.1021/ac202137y (<https://doi.org/10.1021%2Fac202137y>). PMID 22035423 (<https://pubmed.ncbi.nlm.nih.gov/22035423>).

61. Rickel JM, Dixon AJ, Klivanov AL, Hossack JA (August 2018). "A flow focusing microfluidic device with an integrated Coulter particle counter for production, counting and size characterization of monodisperse microbubbles" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6566100>). *Lab on a Chip*. **18** (17): 2653–2664. doi:10.1039/C8LC00496J (<https://doi.org/10.1039/C8LC00496J>). PMC 6566100 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6566100>). PMID 30070301 (<https://pubmed.ncbi.nlm.nih.gov/30070301>).
62. Lewpiriyawong N, Yang C (March 2012). "AC-dielectrophoretic characterization and separation of submicron and micron particles using sidewall AgPDMS electrodes" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3365326>). *Biomicrofluidics*. **6** (1): 12807–128079. doi:10.1063/1.3682049 (<https://doi.org/10.1063/1.3682049>). PMC 3365326 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3365326>). PMID 22662074 (<https://pubmed.ncbi.nlm.nih.gov/22662074>).
63. Gnyawali V, Strohm EM, Wang JZ, Tsai SS, Kolios MC (February 2019). "Simultaneous acoustic and photoacoustic microfluidic flow cytometry for label-free analysis" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6367457>). *Scientific Reports*. **9** (1): 1585. Bibcode:2019NatSR...9.1585G (<https://ui.adsabs.harvard.edu/abs/2019NatSR...9.1585G>). doi:10.1038/s41598-018-37771-5 (<https://doi.org/10.1038/s41598-018-37771-5>). PMC 6367457 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6367457>). PMID 30733497 (<https://pubmed.ncbi.nlm.nih.gov/30733497>).
64. Weiss AC, Krüger K, Besford QA, Schlenk M, Kempe K, Förster S, Caruso F (January 2019). "In Situ Characterization of Protein Corona Formation on Silica Microparticles Using Confocal Laser Scanning Microscopy Combined with Microfluidics". *ACS Applied Materials & Interfaces*. **11** (2): 2459–2469. doi:10.1021/acsami.8b14307 (<https://doi.org/10.1021/acsami.8b14307>). hdl:11343/219876 (<https://hdl.handle.net/11343/219876>). PMID 30600987 (<https://pubmed.ncbi.nlm.nih.gov/30600987>). S2CID 58555221 (<https://api.semanticscholar.org/CorpusID:58555221>).
65. Munaz A, Shiddiky MJ, Nguyen NT (May 2018). "Recent advances and current challenges in magnetophoresis based micro magnetofluidics" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6013300>). *Biomicrofluidics*. **12** (3): 031501. doi:10.1063/1.5035388 (<https://doi.org/10.1063/1.5035388>). PMC 6013300 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6013300>). PMID 29983837 (<https://pubmed.ncbi.nlm.nih.gov/29983837>).
66. Dibaji S, Rezai P (2020-06-01). "Triplex Inertia-Magneto-Elastic (TIME) sorting of microparticles in non-Newtonian fluids". *Journal of Magnetism and Magnetic Materials*. **503**: 166620. Bibcode:2020JMMM..50366620D (<https://ui.adsabs.harvard.edu/abs/2020JMMM..50366620D>). doi:10.1016/j.jmmm.2020.166620 (<https://doi.org/10.1016/j.jmmm.2020.166620>). ISSN 0304-8853 (<https://www.worldcat.org/issn/0304-8853>). S2CID 213233645 (<https://api.semanticscholar.org/CorpusID:213233645>).
67. Alnaimat F, Dagher S, Mathew B, Hilal-Alnqbi A, Khashan S (November 2018). "Microfluidics Based Magnetophoresis: A Review". *Chemical Record*. **18** (11): 1596–1612. doi:10.1002/tcr.201800018 (<https://doi.org/10.1002/tcr.201800018>). PMID 29888856 (<https://pubmed.ncbi.nlm.nih.gov/29888856>). S2CID 47016122 (<https://api.semanticscholar.org/CorpusID:47016122>).
68. Unni M, Zhang J, George TJ, Segal MS, Fan ZH, Rinaldi C (March 2020). "Engineering magnetic nanoparticles and their integration with microfluidics for cell isolation" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7023483>). *Journal of Colloid and Interface Science*. **564**: 204–215. Bibcode:2020JCIS..564..204U (<https://ui.adsabs.harvard.edu/abs/2020JCIS..564..204U>). doi:10.1016/j.jcis.2019.12.092 (<https://doi.org/10.1016/j.jcis.2019.12.092>). PMC 7023483 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7023483>). PMID 31911225 (<https://pubmed.ncbi.nlm.nih.gov/31911225>).

69. Xia N, Hunt TP, Mayers BT, Alsberg E, Whitesides GM, Westervelt RM, Ingber DE (December 2006). "Combined microfluidic-micromagnetic separation of living cells in continuous flow". *Biomedical Microdevices*. **8** (4): 299–308. doi:10.1007/s10544-006-0033-0 (<https://doi.org/10.1007/s10544-006-0033-0>). PMID 17003962 (<https://pubmed.ncbi.nlm.nih.gov/17003962>). S2CID 14534776 (<https://api.semanticscholar.org/CorpusID:14534776>).
70. Pamme N (January 2006). "Magnetism and microfluidics". *Lab on a Chip*. **6** (1): 24–38. doi:10.1039/B513005K (<https://doi.org/10.1039/B513005K>). PMID 16372066 (<https://pubmed.ncbi.nlm.nih.gov/16372066>).
71. Song K, Li G, Zu X, Du Z, Liu L, Hu Z (March 2020). "The Fabrication and Application Mechanism of Microfluidic Systems for High Throughput Biomedical Screening: A Review" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7143183>). *Micromachines*. **11** (3): 297. doi:10.3390/mi11030297 (<https://doi.org/10.3390/mi11030297>). PMC 7143183 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7143183>). PMID 32168977 (<https://pubmed.ncbi.nlm.nih.gov/32168977>).
72. Gao QH, Zhang WM, Zou HX, Li WB, Yan H, Peng ZK, Meng G (2019). "Label-free manipulation via the magneto-Archimedes effect: fundamentals, methodology and applications" (<http://xlink.rsc.org/?DOI=C8MH01616J>). *Materials Horizons*. **6** (7): 1359–1379. doi:10.1039/C8MH01616J (<https://doi.org/10.1039/C8MH01616J>). ISSN 2051-6347 (<http://www.worldcat.org/issn/2051-6347>). S2CID 133309954 (<https://api.semanticscholar.org/CorpusID:133309954>).
73. Akiyama Y, Morishima K (2011-04-18). "Label-free cell aggregate formation based on the magneto-Archimedes effect". *Applied Physics Letters*. **98** (16): 163702. Bibcode:2011ApPhL..98p3702A (<https://ui.adsabs.harvard.edu/abs/2011ApPhL..98p3702A>). doi:10.1063/1.3581883 (<https://doi.org/10.1063/1.3581883>). ISSN 0003-6951 (<http://www.worldcat.org/issn/0003-6951>).
74. Nguyen NT, Wereley S (2006). *Fundamentals and Applications of Microfluidics*. Artech House.
75. DeMello AJ (July 2006). "Control and detection of chemical reactions in microfluidic systems". *Nature*. **442** (7101): 394–402. Bibcode:2006Natur.442..394D (<https://ui.adsabs.harvard.edu/abs/2006Natur.442..394D>). doi:10.1038/nature05062 (<https://doi.org/10.1038/nature05062>). PMID 16871207 (<https://pubmed.ncbi.nlm.nih.gov/16871207>). S2CID 4421580 (<https://api.semanticscholar.org/CorpusID:4421580>).
76. Pawell RS, Inglis DW, Barber TJ, Taylor RA (2013). "Manufacturing and wetting low-cost microfluidic cell separation devices" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3785532>). *Biomicrofluidics*. **7** (5): 56501. doi:10.1063/1.4821315 (<https://doi.org/10.1063/1.4821315>). PMC 3785532 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3785532>). PMID 24404077 (<https://pubmed.ncbi.nlm.nih.gov/24404077>).
77. Pawell RS, Taylor RA, Morris KV, Barber TJ (2015). "Automating microfluidic part verification". *Microfluidics and Nanofluidics*. **18** (4): 657–665. doi:10.1007/s10404-014-1464-1 (<https://doi.org/10.1007/s10404-014-1464-1>). S2CID 96793921 (<https://api.semanticscholar.org/CorpusID:96793921>).
78. Cheng JJ, Nicaise SM, Berggren KK, Gradečak S (January 2016). "Dimensional Tailoring of Hydrothermally Grown Zinc Oxide Nanowire Arrays". *Nano Letters*. **16** (1): 753–759. Bibcode:2016NanoL..16..753C (<https://ui.adsabs.harvard.edu/abs/2016NanoL..16..753C>). doi:10.1021/acs.nanolett.5b04625 (<https://doi.org/10.1021/acs.nanolett.5b04625>). PMID 26708095 (<https://pubmed.ncbi.nlm.nih.gov/26708095>).
79. Herold KE (2009). Rasooly A (ed.). *Lab-on-a-Chip Technology: Fabrication and Microfluidics*. Caister Academic Press. ISBN 978-1-904455-46-2.
80. Herold KE (2009). Rasooly A (ed.). *Lab-on-a-Chip Technology: Biomolecular Separation and Analysis*. Caister Academic Press. ISBN 978-1-904455-47-9.

81. Barrett MP, Cooper JM, Regnault C, Holm SH, Beech JP, Tegenfeldt JO, Hochstetter A (October 2017). "Microfluidics-Based Approaches to the Isolation of African Trypanosomes" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5750571>). *Pathogens*. **6** (4): 47. doi:10.3390/pathogens6040047 (<https://doi.org/10.3390%2Fpathogens6040047>). PMC 5750571 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5750571>). PMID 28981471 (<https://pubmed.ncbi.nlm.nih.gov/28981471>).
82. Jing G, Polaczyk A, Oerther DB, Papautsky I (2007). "Development of a microfluidic biosensor for detection of environmental mycobacteria". *Sensors and Actuators B: Chemical*. **123** (1): 614–621. doi:10.1016/j.snb.2006.07.029 (<https://doi.org/10.1016%2Fj.snb.2006.07.029>).
83. Wang P, Robert L, Pelletier J, Dang WL, Taddei F, Wright A, Jun S (June 2010). "Robust growth of *Escherichia coli*" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2902570>). *Current Biology*. **20** (12): 1099–1103. doi:10.1016/j.cub.2010.04.045 (<https://doi.org/10.1016%2Fj.cub.2010.04.045>). PMC 2902570 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2902570>). PMID 20537537 (<https://pubmed.ncbi.nlm.nih.gov/20537537>).
84. Manbachi A, Shrivastava S, Cioffi M, Chung BG, Moretti M, Demirci U, et al. (May 2008). "Microcirculation within grooved substrates regulates cell positioning and cell docking inside microfluidic channels" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2668874>). *Lab on a Chip*. **8** (5): 747–754. doi:10.1039/B718212K (<https://doi.org/10.1039%2FB718212K>). PMC 2668874 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2668874>). PMID 18432345 (<https://pubmed.ncbi.nlm.nih.gov/18432345>).
85. Yliperttula M, Chung BG, Navaladi A, Manbachi A, Urtti A (October 2008). "High-throughput screening of cell responses to biomaterials". *European Journal of Pharmaceutical Sciences*. **35** (3): 151–160. doi:10.1016/j.ejps.2008.04.012 (<https://doi.org/10.1016%2Fj.ejps.2008.04.012>). PMID 18586092 (<https://pubmed.ncbi.nlm.nih.gov/18586092>).
86. Gilbert DF, Mofrad SA, Friedrich O, Wiest J (February 2019). "Proliferation characteristics of cells cultured under periodic versus static conditions" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6368509>). *Cytotechnology*. **71** (1): 443–452. doi:10.1007/s10616-018-0263-z (<https://doi.org/10.1007%2Fs10616-018-0263-z>). PMC 6368509 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6368509>). PMID 30515656 (<https://pubmed.ncbi.nlm.nih.gov/30515656>).
87. Chung BG, Manbachi A, Saadi W, Lin F, Jeon NL, Khademhosseini A (2007). "A gradient-generating microfluidic device for cell biology" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2565846>). *Journal of Visualized Experiments*. **7** (7): 271. doi:10.3791/271 (<https://doi.org/10.3791%2F271>). PMC 2565846 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2565846>). PMID 18989442 (<https://pubmed.ncbi.nlm.nih.gov/18989442>).
88. Pelletier J, Halvorsen K, Ha BY, Paparcone R, Sandler SJ, Woldringh CL, et al. (October 2012). "Physical manipulation of the *Escherichia coli* chromosome reveals its soft nature" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3479577>). *Proceedings of the National Academy of Sciences of the United States of America*. **109** (40): E2649–E2656. Bibcode:2012PNAS..109E2649P (<https://ui.adsabs.harvard.edu/abs/2012PNAS..109E2649P>). doi:10.1073/pnas.1208689109 (<https://doi.org/10.1073%2Fpnas.1208689109>). PMC 3479577 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3479577>). PMID 22984156 (<https://pubmed.ncbi.nlm.nih.gov/22984156>).
89. Amir A, Babaeipour F, McIntosh DB, Nelson DR, Jun S (April 2014). "Bending forces plastically deform growing bacterial cell walls" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4000856>). *Proceedings of the National Academy of Sciences of the United States of America*. **111** (16): 5778–5783. arXiv:1305.5843 (<https://arxiv.org/abs/1305.5843>). Bibcode:2014PNAS..111.5778A (<https://ui.adsabs.harvard.edu/abs/2014PNAS..111.5778A>). doi:10.1073/pnas.1317497111 (<https://doi.org/10.1073%2Fpnas.1317497111>). PMC 4000856 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4000856>). PMID 24711421 (<https://pubmed.ncbi.nlm.nih.gov/24711421>).

90. Choi JW, Rosset S, Niklaus M, Adleman JR, Shea H, Psaltis D (March 2010). "3-dimensional electrode patterning within a microfluidic channel using metal ion implantation" (<https://authors.library.caltech.edu/17871/>). *Lab on a Chip*. **10** (6): 783–788. doi:10.1039/B917719A (<https://doi.org/10.1039%2FB917719A>). PMID 20221568 (<https://pubmed.ncbi.nlm.nih.gov/20221568>).
91. Yetisen AK, Jiang L, Cooper JR, Qin Y, Palanivelu R, Zohar Y (May 2011). "A microsystem-based assay for studying pollen tube guidance in plant reproduction". *J. Micromech. Microeng.* **25** (5): 054018. Bibcode:2011JMiMi..21e4018Y (<https://ui.adsabs.harvard.edu/abs/2011JMiMi..21e4018Y>). doi:10.1088/0960-1317/21/5/054018 (<https://doi.org/10.1088%2F0960-1317%2F21%2F5%2F054018>). S2CID 12989263 (<https://api.semanticscholar.org/CorpusID:12989263>).
92. Fan H, Das C, Chen H (2009). "Two-Dimensional Electrophoresis in a Chip". In Herold KE, Rasooly A (eds.). *Lab-on-a-Chip Technology: Biomolecular Separation and Analysis*. Caister Academic Press. ISBN 978-1-904455-47-9.
93. Bontoux N, Dauphinot L, Potier MC (2009). "Elaborating Lab-on-a-Chips for Single-cell Transcriptome Analysis". In Herold KE, Rasooly A (eds.). *Lab-on-a-Chip Technology: Biomolecular Separation and Analysis*. Caister Academic Press. ISBN 978-1-904455-47-9.
94. Cady NC (2009). "Microchip-based PCR Amplification Systems". *Lab-on-a-Chip Technology: Biomolecular Separation and Analysis*. Caister Academic Press. ISBN 978-1-904455-47-9.
95. Keymer JE, Galajda P, Muldoon C, Park S, Austin RH (November 2006). "Bacterial metapopulations in nanofabricated landscapes" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1635019>). *Proceedings of the National Academy of Sciences of the United States of America*. **103** (46): 17290–17295. Bibcode:2006PNAS..10317290K (<https://ui.adsabs.harvard.edu/abs/2006PNAS..10317290K>). doi:10.1073/pnas.0607971103 (<https://doi.org/10.1073%2Fpnas.0607971103>). PMC 1635019 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1635019>). PMID 17090676 (<https://pubmed.ncbi.nlm.nih.gov/17090676>).
96. Hochstetter A, Stellamanns E, Deshpande S, Uppaluri S, Engstler M, Pfohl T (April 2015). "Microfluidics-based single cell analysis reveals drug-dependent motility changes in trypanosomes" (<https://edoc.unibas.ch/41485/1/C5LC00124B.pdf>) (PDF). *Lab on a Chip*. **15** (8): 1961–1968. doi:10.1039/C5LC00124B (<https://doi.org/10.1039%2FC5LC00124B>). PMID 25756872 (<https://pubmed.ncbi.nlm.nih.gov/25756872>).
97. Ahmed T, Shimizu TS, Stocker R (November 2010). "Microfluidics for bacterial chemotaxis". *Integrative Biology*. **2** (11–12): 604–629. doi:10.1039/C0IB00049C (<https://doi.org/10.1039%2FC0IB00049C>). hdl:1721.1/66851 (<https://hdl.handle.net/1721.1%2F66851>). PMID 20967322 (<https://pubmed.ncbi.nlm.nih.gov/20967322>).
98. Seymour JR, Simó R, Ahmed T, Stocker R (July 2010). "Chemoattraction to dimethylsulfoniopropionate throughout the marine microbial food web". *Science*. **329** (5989): 342–345. Bibcode:2010Sci...329..342S (<https://ui.adsabs.harvard.edu/abs/2010Sci...329..342S>). doi:10.1126/science.1188418 (<https://doi.org/10.1126%2Fscience.1188418>). PMID 20647471 (<https://pubmed.ncbi.nlm.nih.gov/20647471>). S2CID 12511973 (<https://api.semanticscholar.org/CorpusID:12511973>).
99. Galajda P, Keymer J, Chaikin P, Austin R (December 2007). "A wall of funnels concentrates swimming bacteria" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2168927>). *Journal of Bacteriology*. **189** (23): 8704–8707. doi:10.1128/JB.01033-07 (<https://doi.org/10.1128%2FJB.01033-07>). PMC 2168927 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2168927>). PMID 17890308 (<https://pubmed.ncbi.nlm.nih.gov/17890308>).

100. Angelani L, Di Leonardo R, Ruocco G (January 2009). "Self-starting micromotors in a bacterial bath". *Physical Review Letters*. **102** (4): 048104. arXiv:0812.2375 (<https://arxiv.org/abs/0812.2375>). Bibcode:2009PhRvL.102d8104A (<https://ui.adsabs.harvard.edu/abs/2009PhRvL.102d8104A>). doi:10.1103/PhysRevLett.102.048104 (<https://doi.org/10.1103%2FPhysRevLett.102.048104>). PMID 19257480 (<https://pubmed.ncbi.nlm.nih.gov/19257480>). S2CID 33943502 (<https://api.semanticscholar.org/CorpusID:33943502>).
101. Di Leonardo R, Angelani L, Dell'arciprete D, Ruocco G, Iebba V, Schippa S, et al. (May 2010). "Bacterial ratchet motors" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2906854>). *Proceedings of the National Academy of Sciences of the United States of America*. **107** (21): 9541–9545. arXiv:0910.2899 (<https://arxiv.org/abs/0910.2899>). Bibcode:2010PNAS..107.9541D (<https://ui.adsabs.harvard.edu/abs/2010PNAS..107.9541D>). doi:10.1073/pnas.0910426107 (<https://doi.org/10.1073%2Fpnas.0910426107>). PMC 2906854 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2906854>). PMID 20457936 (<https://pubmed.ncbi.nlm.nih.gov/20457936>).
102. Sokolov A, Apodaca MM, Grzybowski BA, Aranson IS (January 2010). "Swimming bacteria power microscopic gears" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2824308>). *Proceedings of the National Academy of Sciences of the United States of America*. **107** (3): 969–974. Bibcode:2010PNAS..107..969S (<https://ui.adsabs.harvard.edu/abs/2010PNAS..107..969S>). doi:10.1073/pnas.0913015107 (<https://doi.org/10.1073%2Fpnas.0913015107>). PMC 2824308 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2824308>). PMID 20080560 (<https://pubmed.ncbi.nlm.nih.gov/20080560>).
103. Grilli S, Miccio L, Vespini V, Finizio A, De Nicola S, Ferraro P (May 2008). "Liquid micro-lens array activated by selective electrowetting on lithium niobate substrates" (<https://doi.org/10.1364%2FOE.16.008084>). *Optics Express*. **16** (11): 8084–8093. Bibcode:2008OExpr..16.8084G (<https://ui.adsabs.harvard.edu/abs/2008OExpr..16.8084G>). doi:10.1364/OE.16.008084 (<https://doi.org/10.1364%2FOE.16.008084>). PMID 18545521 (<https://pubmed.ncbi.nlm.nih.gov/18545521>). S2CID 15923737 (<https://api.semanticscholar.org/CorpusID:15923737>).
104. Ferraro P, Miccio L, Grilli S, Finizio A, De Nicola S, Vespini V (2008). "Manipulating Thin Liquid Films for Tunable Microlens Arrays". *Optics and Photonics News*. **19** (12): 34. doi:10.1364/OPN.19.12.000034 (<https://doi.org/10.1364%2FOPN.19.12.000034>).
105. Pégard NC, Toth ML, Driscoll M, Fleischer JW (December 2014). "Flow-scanning optical tomography" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5859944>). *Lab on a Chip*. **14** (23): 4447–4450. doi:10.1039/C4LC00701H (<https://doi.org/10.1039%2FC4LC00701H>). PMC 5859944 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5859944>). PMID 25256716 (<https://pubmed.ncbi.nlm.nih.gov/25256716>).
106. Pégard NC, Fleischer JW (2012). "3D microfluidic microscopy using a tilted channel". *Biomedical Optics and 3-D Imaging*. pp. BM4B.4. doi:10.1364/BIOMED.2012.BM4B.4 (<https://doi.org/10.1364%2FBIOMED.2012.BM4B.4>). ISBN 978-1-55752-942-8.
107. Lu CH, Pégard NC, Fleischer JW (22 April 2013). "Flow-based structured illumination". *Applied Physics Letters*. **102** (16): 161115. Bibcode:2013ApPhL.102p1115L (<https://ui.adsabs.harvard.edu/abs/2013ApPhL.102p1115L>). doi:10.1063/1.4802091 (<https://doi.org/10.1063%2F1.4802091>).
108. Kirsanov, D.; Babain, V.; Agafonova-Moroz, M.; Lumpov, A.; Legin, A. (2012-03-01). "Combination of optical spectroscopy and chemometric techniques—a possible way for on-line monitoring of spent nuclear fuel (SNF) reprocessing" (<https://www.degruyter.com/document/doi/10.1524/ract.2012.1901/html>). *Radiochimica Acta*. **100** (3): 185–188. doi:10.1524/ract.2012.1901 (<https://doi.org/10.1524%2Fract.2012.1901>). S2CID 101475605 (<https://api.semanticscholar.org/CorpusID:101475605>).

109. Nelson, Gilbert L.; Lackey, Hope E.; Bello, Job M.; Felmy, Heather M.; Bryan, Hannah B.; Lamadie, Fabrice; Bryan, Samuel A.; Lines, Amanda M. (2021-01-26). "Enabling Microscale Processing: Combined Raman and Absorbance Spectroscopy for Microfluidic On-Line Monitoring" (<https://pubs.acs.org/doi/10.1021/acs.analchem.0c04225>). *Analytical Chemistry*. **93** (3): 1643–1651. doi:10.1021/acs.analchem.0c04225 (<https://doi.org/10.1021%2Facs.analchem.0c04225>). ISSN 0003-2700 (<https://www.worldcat.org/issn/0003-2700>). OSTI 1783814 (<https://www.osti.gov/biblio/1783814>). PMID 33337856 (<https://pubmed.ncbi.nlm.nih.gov/33337856>). S2CID 229323758 (<https://api.semanticscholar.org/CorpusID:229323758>).
110. Mattio, Elodie; Caleyron, Audrey; Miguirditchian, Manuel; Lines, Amanda M.; Bryan, Samuel A.; Lackey, Hope E.; Rodriguez-Ruiz, Isaac; Lamadie, Fabrice (May 2022). "Microfluidic In-Situ Spectrophotometric Approaches to Tackle Actinides Analysis in Multiple Oxidation States" (<http://journals.sagepub.com/doi/10.1177/00037028211063916>). *Applied Spectroscopy*. **76** (5): 580–589. Bibcode:2022ApSpe..76..580M (<https://ui.adsabs.harvard.edu/abs/2022ApSpe..76..580M>). doi:10.1177/00037028211063916 (<https://doi.org/10.1177%2F00037028211063916>). ISSN 0003-7028 (<https://www.worldcat.org/issn/0003-7028>). PMID 35108115 (<https://pubmed.ncbi.nlm.nih.gov/35108115>). S2CID 246488502 (<https://api.semanticscholar.org/CorpusID:246488502>) – via Sage Journals.
111. Bryan, S. A.; Levitskaia, Tatiana G.; Johnsen, A. M.; Orton, C. R.; Peterson, J. M. (September 2011). "Spectroscopic monitoring of spent nuclear fuel reprocessing streams: an evaluation of spent fuel solutions via Raman, visible, and near-infrared spectroscopy" (<https://www.degruyter.com/document/doi/10.1524/ract.2011.1865/html>). *Radiochimica Acta*. **99** (9): 563–572. doi:10.1524/ract.2011.1865 (<https://doi.org/10.1524%2Fract.2011.1865>). ISSN 0033-8230 (<https://www.worldcat.org/issn/0033-8230>). S2CID 95632074 (<https://api.semanticscholar.org/CorpusID:95632074>).
112. Nelson, Gilbert L.; Lines, Amanda M.; Bello, Job M.; Bryan, Samuel A. (2019-09-27). "Online Monitoring of Solutions Within Microfluidic Chips: Simultaneous Raman and UV–Vis Absorption Spectroscopies" (<https://pubs.acs.org/doi/10.1021/acssensors.9b00736>). *ACS Sensors*. **4** (9): 2288–2295. doi:10.1021/acssensors.9b00736 (<https://doi.org/10.1021%2Facssensors.9b00736>). ISSN 2379-3694 (<https://www.worldcat.org/issn/2379-3694>). PMID 31434479 (<https://pubmed.ncbi.nlm.nih.gov/31434479>). S2CID 201275176 (<https://api.semanticscholar.org/CorpusID:201275176>).
113. Rodríguez-Ruiz, Isaac; Lamadie, Fabrice; Charton, Sophie (2018-02-20). "Uranium(VI) On-Chip Microliter Concentration Measurements in a Highly Extended UV–Visible Absorbance Linearity Range" (<https://pubs.acs.org/doi/10.1021/acs.analchem.7b05162>). *Analytical Chemistry*. **90** (4): 2456–2460. doi:10.1021/acs.analchem.7b05162 (<https://doi.org/10.1021%2Facs.analchem.7b05162>). ISSN 0003-2700 (<https://www.worldcat.org/issn/0003-2700>). PMID 29327582 (<https://pubmed.ncbi.nlm.nih.gov/29327582>).
114. Mattio, Elodie; Lamadie, Fabrice; Rodriguez-Ruiz, Isaac; Cames, Beatrice; Charton, Sophie (2020-02-01). "Photonic Lab-on-a-Chip analytical systems for nuclear applications: optical performance and UV–Vis–IR material characterization after chemical exposure and gamma irradiation" (<https://doi.org/10.1007/s10967-019-06992-x>). *Journal of Radioanalytical and Nuclear Chemistry*. **323** (2): 965–973. doi:10.1007/s10967-019-06992-x (<https://doi.org/10.1007%2Fs10967-019-06992-x>). ISSN 1588-2780 (<https://www.worldcat.org/issn/1588-2780>). S2CID 209441127 (<https://api.semanticscholar.org/CorpusID:209441127>).
115. Kim JY, Cho SW, Kang DK, Edel JB, Chang SI, deMello AJ, O'Hare D (September 2012). "Lab-chip HPLC with integrated droplet-based microfluidics for separation and high frequency compartmentalisation". *Chemical Communications*. **48** (73): 9144–9146. doi:10.1039/c2cc33774f (<https://doi.org/10.1039%2Fc2cc33774f>). PMID 22871959 (<https://pubmed.ncbi.nlm.nih.gov/22871959>).

116. Ochoa A, Álvarez-Bohórquez E, Castellero E, Olguin LF (May 2017). "Detection of Enzyme Inhibitors in Crude Natural Extracts Using Droplet-Based Microfluidics Coupled to HPLC". *Analytical Chemistry*. **89** (9): 4889–4896. doi:10.1021/acs.analchem.6b04988 (<https://doi.org/10.1021%2Facs.analchem.6b04988>). PMID 28374582 (<https://pubmed.ncbi.nlm.nih.gov/28374582>).
117. Gerhardt RF, Peretzki AJ, Piendl SK, Belder D (December 2017). "Seamless Combination of High-Pressure Chip-HPLC and Droplet Microfluidics on an Integrated Microfluidic Glass Chip". *Analytical Chemistry*. **89** (23): 13030–13037. doi:10.1021/acs.analchem.7b04331 (<https://doi.org/10.1021%2Facs.analchem.7b04331>). PMID 29096060 (<https://pubmed.ncbi.nlm.nih.gov/29096060>).
118. Killeen K, Yin H, Sobek D, Brennen R, Van de Goor T (October 2003). *Chip-LC/MS: HPLC-MS using polymer microfluidics* (<https://www.rsc.org/binaries/LOC/2003/Volume1/120-376.pdf>) (PDF). 7th International Conference on Miniaturized Chemical and Biochemical Analysis Systems. *Proc MicroTAS*. Squaw Valley, California USA. pp. 481–484.
119. Vollmer M, Hörth P, Rozing G, Couté Y, Grimm R, Hochstrasser D, Sanchez JC (March 2006). "Multi-dimensional HPLC/MS of the nucleolar proteome using HPLC-chip/MS". *Journal of Separation Science*. **29** (4): 499–509. doi:10.1002/jssc.200500334 (<https://doi.org/10.1002%2Fjssc.200500334>). PMID 16583688 (<https://pubmed.ncbi.nlm.nih.gov/16583688>).
120. Reichmuth DS, Shepodd TJ, Kirby BJ (May 2005). "Microchip HPLC of peptides and proteins". *Analytical Chemistry*. **77** (9): 2997–3000. doi:10.1021/ac048358r (<https://doi.org/10.1021%2Fac048358r>). PMID 15859622 (<https://pubmed.ncbi.nlm.nih.gov/15859622>).
121. Hardouin J, Duchateau M, Joubert-Caron R, Caron M (2006). "Usefulness of an integrated microfluidic device (HPLC-Chip-MS) to enhance confidence in protein identification by proteomics". *Rapid Communications in Mass Spectrometry*. **20** (21): 3236–3244. Bibcode:2006RCMS...20.3236H (<https://ui.adsabs.harvard.edu/abs/2006RCMS...20.3236H>). doi:10.1002/rcm.2725 (<https://doi.org/10.1002%2Frcm.2725>). PMID 17016832 (<https://pubmed.ncbi.nlm.nih.gov/17016832>).
122. Brennen RA, Yin H, Killeen KP (December 2007). "Microfluidic gradient formation for nanoflow chip LC". *Analytical Chemistry*. **79** (24): 9302–9309. doi:10.1021/ac0712805 (<https://doi.org/10.1021%2Fac0712805>). PMID 17997523 (<https://pubmed.ncbi.nlm.nih.gov/17997523>).
123. Zhu KY, Leung KW, Ting AK, Wong ZC, Ng WY, Choi RC, et al. (March 2012). "Microfluidic chip based nano liquid chromatography coupled to tandem mass spectrometry for the determination of abused drugs and metabolites in human hair". *Analytical and Bioanalytical Chemistry*. **402** (9): 2805–2815. doi:10.1007/s00216-012-5711-6 (<https://doi.org/10.1007%2Fs00216-012-5711-6>). PMID 22281681 (<https://pubmed.ncbi.nlm.nih.gov/22281681>). S2CID 7748546 (<https://api.semanticscholar.org/CorpusID:7748546>).
124. Polat AN, Kraiczek K, Heck AJ, Rajmakers R, Mohammed S (November 2012). "Fully automated isotopic dimethyl labeling and phosphopeptide enrichment using a microfluidic HPLC phosphochip". *Analytical and Bioanalytical Chemistry*. **404** (8): 2507–2512. doi:10.1007/s00216-012-6395-7 (<https://doi.org/10.1007%2Fs00216-012-6395-7>). PMID 22975804 (<https://pubmed.ncbi.nlm.nih.gov/22975804>). S2CID 32545802 (<https://api.semanticscholar.org/CorpusID:32545802>).
125. Santiago JG. "Water Management in PEM Fuel Cells" (https://web.archive.org/web/20080628192632/http://microfluidics.stanford.edu/fuel_cells.htm). *Stanford Microfluidics Laboratory*. Archived from the original (http://microfluidics.stanford.edu/fuel_cells.htm) on 28 June 2008.
126. Tretkoff E (May 2005). "Building a Better Fuel Cell Using Microfluidics" (<http://www.aps.org/publications/apsnews/200505/fuel.cfm>). *APS News*. **14** (5): 3.
127. Allen J. "Fuel Cell Initiative at MnIT Microfluidics Laboratory" (<https://web.archive.org/web/20080305215325/http://www.me.mtu.edu/mnit/>). Michigan Technological University. Archived from the original (<http://www.me.mtu.edu/mnit/>) on 2008-03-05.

128. "NASA Astrobiology Strategy, 2015" (https://web.archive.org/web/20161222190306/https://nai.nasa.gov/media/medialibrary/2015/10/NASA_Astrobiology_Strategy_2015_151008.pdf) (PDF). Archived from the original (https://nai.nasa.gov/media/medialibrary/2015/10/NASA_Astrobiology_Strategy_2015_151008.pdf) (PDF) on 2016-12-22.
129. Beebe DJ, Mensing GA, Walker GM (2002). "Physics and applications of microfluidics in biology". *Annual Review of Biomedical Engineering*. **4**: 261–286. doi:10.1146/annurev.bioeng.4.112601.125916 (<https://doi.org/10.1146/annurev.bioeng.4.112601.125916>). PMID 12117759 (<https://pubmed.ncbi.nlm.nih.gov/12117759/>).
130. Theberge AB, Courtois F, Schaerli Y, Fischlechner M, Abell C, Hollfelder F, Huck WT (August 2010). "Microdroplets in microfluidics: an evolving platform for discoveries in chemistry and biology" (https://eprints.soton.ac.uk/340192/2/_soton.ac.uk_ude_personalfiles_users_jks1m11_mydesktop_theberge_2010_microdroplets%2520in%2520microfluidics-an%2520evolving%2520platform%2520for%2520discoveries%2520in%2520chemistry%2520and%2520biology.pdf) (PDF). *Angewandte Chemie*. **49** (34): 5846–5868. doi:10.1002/anie.200906653 (<https://doi.org/10.1002/anie.200906653>). PMID 20572214 (<https://pubmed.ncbi.nlm.nih.gov/20572214/>). S2CID 18609389 (<https://api.semanticscholar.org/CorpusID:18609389>).
131. van Dinter AM, Schroën CG, Vergeldt FJ, van der Sman RG, Boom RM (May 2012). "Suspension flow in microfluidic devices--a review of experimental techniques focussing on concentration and velocity gradients". *Advances in Colloid and Interface Science*. **173**: 23–34. doi:10.1016/j.cis.2012.02.003 (<https://doi.org/10.1016/j.cis.2012.02.003>). PMID 22405541 (<https://pubmed.ncbi.nlm.nih.gov/22405541/>).
132. Mora MF, Greer F, Stockton AM, Bryant S, Willis PA (November 2011). "Toward total automation of microfluidics for extraterrestrial [sic] in situ analysis". *Analytical Chemistry*. **83** (22): 8636–8641. doi:10.1021/ac202095k (<https://doi.org/10.1021/ac202095k>). PMID 21972965 (<https://pubmed.ncbi.nlm.nih.gov/21972965/>).
133. Chiesl TN, Chu WK, Stockton AM, Amashukeli X, Grunthaner F, Mathies RA (April 2009). "Enhanced amine and amino acid analysis using Pacific Blue and the Mars Organic Analyzer microchip capillary electrophoresis system". *Analytical Chemistry*. **81** (7): 2537–2544. doi:10.1021/ac8023334 (<https://doi.org/10.1021/ac8023334>). PMID 19245228 (<https://pubmed.ncbi.nlm.nih.gov/19245228/>).
134. Kaiser RI, Stockton AM, Kim YS, Jensen EC, Mathies RA (2013). "On the Formation of Dipeptides in Interstellar Model Ices" (<https://doi.org/10.1088/0004-637X/765/2/111>). *The Astrophysical Journal*. **765** (2): 111. Bibcode:2013ApJ...765..111K (<https://ui.adsabs.harvard.edu/abs/2013ApJ...765..111K>). doi:10.1088/0004-637X/765/2/111 (<https://doi.org/10.1088/0004-637X/765/2/111>). ISSN 0004-637X (<https://www.worldcat.org/issn/0004-637X>). S2CID 45120615 (<https://api.semanticscholar.org/CorpusID:45120615>).
135. Stockton AM, Tjin CC, Chiesl TN, Mathies RA (July 2011). "Analysis of carbonaceous biomarkers with the Mars Organic Analyzer microchip capillary electrophoresis system: carboxylic acids". *Astrobiology*. **11** (6): 519–528. Bibcode:2011AsBio..11..519S (<https://ui.adsabs.harvard.edu/abs/2011AsBio..11..519S>). doi:10.1089/ast.2011.0634 (<https://doi.org/10.1089/ast.2011.0634>). PMID 21790324 (<https://pubmed.ncbi.nlm.nih.gov/21790324/>).
136. Stockton AM, Tjin CC, Huang GL, Benhabib M, Chiesl TN, Mathies RA (November 2010). "Analysis of carbonaceous biomarkers with the Mars Organic Analyzer microchip capillary electrophoresis system: aldehydes and ketones". *Electrophoresis*. **31** (22): 3642–3649. doi:10.1002/elps.201000424 (<https://doi.org/10.1002/elps.201000424>). PMID 20967779 (<https://pubmed.ncbi.nlm.nih.gov/20967779/>). S2CID 34503284 (<https://api.semanticscholar.org/CorpusID:34503284>).

137. Mora MF, Stockton AM, Willis PA (2015). "Analysis of thiols by microchip capillary electrophoresis for in situ planetary investigations". *Microchip Capillary Electrophoresis Protocols*. Methods in Molecular Biology. Vol. 1274. New York, NY: Humana Press. pp. 43–52. doi:10.1007/978-1-4939-2353-3_4 (https://doi.org/10.1007%2F978-1-4939-2353-3_4). ISBN 9781493923526. PMID 25673481 (<https://pubmed.ncbi.nlm.nih.gov/25673481>).
138. Bowden SA, Wilson R, Taylor C, Cooper JM, Parnell J (January 2007). "The extraction of intracrystalline biomarkers and other organic compounds from sulphate minerals using a microfluidic format – a feasibility study for remote fossil-life detection using a microfluidic H-cell" (<https://www.cambridge.org/core/journals/international-journal-of-astrobiology/article/the-extraction-of-intracrystalline-biomarkers-and-other-organic-compounds-from-sulphate-minerals-using-a-microfluidic-format-a-feasibility-study-for-remote-fossil-life-detection-using-a-microfluidic-h-cell/C66EB723687CB09F2DAB638A074CBA9D>). *International Journal of Astrobiology*. **6** (1): 27–36. Bibcode:2007IJAsB...6...27B (<https://ui.adsabs.harvard.edu/abs/2007IJAsB...6...27B>). doi:10.1017/S147355040600351X (<https://doi.org/10.1017%2FS147355040600351X>). ISSN 1475-3006 (<https://www.worldcat.org/issn/1475-3006>). S2CID 123048038 (<https://api.semanticscholar.org/CorpusID:123048038>).
139. Neethirajan, Suresh; Kobayashi, Isao; Nakajima, Mitsutoshi; Wu, Dan; Nandagopal, Saravanan; Lin, Francis (2011). "Microfluidics for food, agriculture and biosystems industries" (<http://xlink.rsc.org/?DOI=c0lc00230e>). *Lab on a Chip*. **11** (9): 1574–1586. doi:10.1039/c0lc00230e (<https://doi.org/10.1039%2Fc0lc00230e>). ISSN 1473-0197 (<https://www.worldcat.org/issn/1473-0197>). PMID 21431239 (<https://pubmed.ncbi.nlm.nih.gov/21431239>).
140. Verma, Kiran; Tarafdar, Ayon; Badgujar, Prarabdh C. (January 2021). "Microfluidics assisted tragacanth gum based sub-micron curcumin suspension and its characterization" (<https://dx.doi.org/10.1016/j.lwt.2020.110269>). *LWT*. **135**: 110269. doi:10.1016/j.lwt.2020.110269 (<https://doi.org/10.1016%2Fj.lwt.2020.110269>). ISSN 0023-6438 (<https://www.worldcat.org/issn/0023-6438>). S2CID 224875232 (<https://api.semanticscholar.org/CorpusID:224875232>).
141. Hsiao, Ching-Ju; Lin, Jui-Fen; Wen, Hsin-Yi; Lin, Yu-Mei; Yang, Chih-Hui; Huang, Keng-Shiang; Shaw, Jei-Fu (2020-02-15). "Enhancement of the stability of chlorophyll using chlorophyll-encapsulated polycaprolactone microparticles based on droplet microfluidics" (<https://www.sciencedirect.com/science/article/pii/S0308814619314116>). *Food Chemistry*. **306**: 125300. doi:10.1016/j.foodchem.2019.125300 (<https://doi.org/10.1016%2Fj.foodchem.2019.125300>). ISSN 0308-8146 (<https://www.worldcat.org/issn/0308-8146>). PMID 31562927 (<https://pubmed.ncbi.nlm.nih.gov/31562927>). S2CID 201219877 (<https://api.semanticscholar.org/CorpusID:201219877>).
142. He, Shan; Joseph, Nikita; Feng, Shilun; Jellicoe, Matt; Raston, Colin L. (2020). "Application of microfluidic technology in food processing" (<https://dx.doi.org/10.1039/d0fo01278e>). *Food & Function*. **11** (7): 5726–5737. doi:10.1039/d0fo01278e (<https://doi.org/10.1039%2Fd0fo01278e>). ISSN 2042-6496 (<https://www.worldcat.org/issn/2042-6496>). PMID 32584365 (<https://pubmed.ncbi.nlm.nih.gov/32584365>). S2CID 220059922 (<https://api.semanticscholar.org/CorpusID:220059922>).
143. Hinderink, Emma B. A.; Kaade, Wael; Sagis, Leonard; Schroën, Karin; Berton-Carabin, Claire C. (2020-05-01). "Microfluidic investigation of the coalescence susceptibility of pea protein-stabilised emulsions: Effect of protein oxidation level" (<https://www.sciencedirect.com/science/article/pii/S0268005X19317345>). *Food Hydrocolloids*. **102**: 105610. doi:10.1016/j.foodhyd.2019.105610 (<https://doi.org/10.1016%2Fj.foodhyd.2019.105610>). ISSN 0268-005X (<https://www.worldcat.org/issn/0268-005X>). S2CID 212935489 (<https://api.semanticscholar.org/CorpusID:212935489>).

144. Zhang, Jia; Xu, Wenhua; Xu, Fengying; Lu, Wangwang; Hu, Liuyun; Zhou, Jianlin; Zhang, Chen; Jiang, Zhuo (February 2021). "Microfluidic droplet formation in co-flow devices fabricated by micro 3D printing" (<https://dx.doi.org/10.1016/j.jfoodeng.2020.110212>). *Journal of Food Engineering*. **290**: 110212. doi:10.1016/j.jfoodeng.2020.110212 (<https://doi.org/10.1016%2Fj.jfoodeng.2020.110212>). ISSN 0260-8774 (<https://www.worldcat.org/issn/0260-8774>). S2CID 224841971 (<https://api.semanticscholar.org/CorpusID:224841971>).
145. Harmon JB, Gray HK, Young CC, Schwab KJ (2020) Microfluidic droplet application for bacterial surveillance in fresh-cut produce wash waters. *PLoS ONE* 15(6): e0233239. <https://doi.org/10.1371/journal.pone.0233239>
146. Trofimchuk, Evan; Hu, Yaxi; Nilghaz, Azadeh; Hua, Marti Z.; Sun, Selina; Lu, Xiaonan (2020-06-30). "Development of paper-based microfluidic device for the determination of nitrite in meat" (<https://www.sciencedirect.com/science/article/pii/S0308814620302557>). *Food Chemistry*. **316**: 126396. doi:10.1016/j.foodchem.2020.126396 (<https://doi.org/10.1016%2Fj.foodchem.2020.126396>). ISSN 0308-8146 (<https://www.worldcat.org/issn/0308-8146>). PMID 32066068 (<https://pubmed.ncbi.nlm.nih.gov/32066068>). S2CID 211160645 (<https://api.semanticscholar.org/CorpusID:211160645>).
147. Ko, Chien-Hsuan; Liu, Chan-Chiung; Chen, Kuan-Hong; Sheu, Fuu; Fu, Lung-Ming; Chen, Szu-Jui (2021-05-30). "Microfluidic colorimetric analysis system for sodium benzoate detection in foods" (<https://www.sciencedirect.com/science/article/pii/S0308814620326352>). *Food Chemistry*. **345**: 128773. doi:10.1016/j.foodchem.2020.128773 (<https://doi.org/10.1016%2Fj.foodchem.2020.128773>). ISSN 0308-8146 (<https://www.worldcat.org/issn/0308-8146>). PMID 33302108 (<https://pubmed.ncbi.nlm.nih.gov/33302108>). S2CID 228100279 (<https://api.semanticscholar.org/CorpusID:228100279>).
148. Trofimchuk, Evan; Nilghaz, Azadeh; Sun, Selina; Lu, Xiaonan (2020). "Determination of norfloxacin residues in foods by exploiting the coffee-ring effect and paper-based microfluidics device coupling with smartphone-based detection" (<https://onlinelibrary.wiley.com/doi/abs/10.1111/1750-3841.15039>). *Journal of Food Science*. **85** (3): 736–743. doi:10.1111/1750-3841.15039 (<https://doi.org/10.1111%2F1750-3841.15039>). ISSN 1750-3841 (<https://www.worldcat.org/issn/1750-3841>). PMID 32017096 (<https://pubmed.ncbi.nlm.nih.gov/32017096>). S2CID 211023292 (<https://api.semanticscholar.org/CorpusID:211023292>).
149. Hajji I, Serra M, Geremie L, Ferrante I, Renault R, Viovy JL, Descroix S, Ferraro D (2020). "Droplet microfluidic platform for fast and continuous-flow RT-qPCR analysis devoted to cancer diagnosis application". *Sensors and Actuators B: Chemical*. **303**: 127171. doi:10.1016/j.snb.2019.127171 (<https://doi.org/10.1016%2Fj.snb.2019.127171>). S2CID 208705450 (<https://api.semanticscholar.org/CorpusID:208705450>).
150. Macosko EZ, Basu A, Satija R, Nemesh J, Shekhar K, Goldman M, et al. (May 2015). "Highly Parallel Genome-wide Expression Profiling of Individual Cells Using Nanoliter Droplets" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4481139>). *Cell*. **161** (5): 1202–1214. doi:10.1016/j.cell.2015.05.002 (<https://doi.org/10.1016%2Fj.cell.2015.05.002>). PMC 4481139 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4481139>). PMID 26000488 (<https://pubmed.ncbi.nlm.nih.gov/26000488>).
151. Liu P, Liang H, Xue L, Yang C, Liu Y, Zhou K, Jiang X (July 2012). "Potential clinical significance of plasma-based KRAS mutation analysis using the COLD-PCR/TaqMan® - MGB probe genotyping method" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3460285>). *Experimental and Therapeutic Medicine*. **4** (1): 109–112. doi:10.3892/etm.2012.566 (<https://doi.org/10.3892%2Fetm.2012.566>). PMC 3460285 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3460285>). PMID 23060932 (<https://pubmed.ncbi.nlm.nih.gov/23060932>).

152. Manak MS, Varsanik JS, Hogan BJ, Whitfield MJ, Su WR, Joshi N, et al. (October 2018). "Live-cell phenotypic-biomarker microfluidic assay for the risk stratification of cancer patients via machine learning" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6407716>). *Nature Biomedical Engineering*. **2** (10): 761–772. doi:10.1038/s41551-018-0285-z (<https://doi.org/10.1038/s41551-018-0285-z>). PMC 6407716 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6407716>). PMID 30854249 (<https://pubmed.ncbi.nlm.nih.gov/30854249>).
153. Karabacak NM, Spuhler PS, Fachin F, Lim EJ, Pai V, Ozkumur E, et al. (March 2014). "Microfluidic, marker-free isolation of circulating tumor cells from blood samples" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4179254>). *Nature Protocols*. **9** (3): 694–710. doi:10.1038/nprot.2014.044 (<https://doi.org/10.1038/nprot.2014.044>). PMC 4179254 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4179254>). PMID 24577360 (<https://pubmed.ncbi.nlm.nih.gov/24577360>).
154. Warburg O, Wind F, Negelein E (March 1927). "The Metabolism of Tumors in the Body" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2140820>). *The Journal of General Physiology*. **8** (6): 519–530. doi:10.1085/jgp.8.6.519 (<https://doi.org/10.1085/jgp.8.6.519>). PMC 2140820 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2140820>). PMID 19872213 (<https://pubmed.ncbi.nlm.nih.gov/19872213>).
155. Gascoyne PR, Noshari J, Anderson TJ, Becker FF (April 2009). "Isolation of rare cells from cell mixtures by dielectrophoresis" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3754902>). *Electrophoresis*. **30** (8): 1388–1398. doi:10.1002/elps.200800373 (<https://doi.org/10.1002/elps.200800373>). PMC 3754902 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3754902>). PMID 19306266 (<https://pubmed.ncbi.nlm.nih.gov/19306266>).
156. Yu M, Bardia A, Aceto N, Bersani F, Madden MW, Donaldson MC, et al. (July 2014). "Cancer therapy. Ex vivo culture of circulating breast tumor cells for individualized testing of drug susceptibility" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4358808>). *Science*. **345** (6193): 216–220. Bibcode:2014Sci...345..216Y (<https://ui.adsabs.harvard.edu/abs/2014Sci...345..216Y>). doi:10.1126/science.1253533 (<https://doi.org/10.1126/science.1253533>). PMC 4358808 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4358808>). PMID 25013076 (<https://pubmed.ncbi.nlm.nih.gov/25013076>).
157. Liang LG, Kong MQ, Zhou S, Sheng YF, Wang P, Yu T, et al. (April 2017). "An integrated double-filtration microfluidic device for isolation, enrichment and quantification of urinary extracellular vesicles for detection of bladder cancer" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5402302>). *Scientific Reports*. **7** (1): 46224. Bibcode:2017NatSR...746224L (<https://ui.adsabs.harvard.edu/abs/2017NatSR...746224L>). doi:10.1038/srep46224 (<https://doi.org/10.1038/srep46224>). PMC 5402302 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5402302>). PMID 28436447 (<https://pubmed.ncbi.nlm.nih.gov/28436447>).
158. Mathew DG, Beekman P, Lemay SG, Zuilhof H, Le Gac S, van der Wiel WG (February 2020). "Electrochemical Detection of Tumor-Derived Extracellular Vesicles on Nanointerdigitated Electrodes" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7020140>). *Nano Letters*. **20** (2): 820–828. Bibcode:2020NanoL..20..820M (<https://ui.adsabs.harvard.edu/abs/2020NanoL..20..820M>). doi:10.1021/acs.nanolett.9b02741 (<https://doi.org/10.1021/acs.nanolett.9b02741>). PMC 7020140 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7020140>). PMID 31536360 (<https://pubmed.ncbi.nlm.nih.gov/31536360>).
159. Liu Z, Lee Y, Jang JH, Li Y, Han X, Yokoi K, et al. (September 2015). "Microfluidic cytometric analysis of cancer cell transportability and invasiveness" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4585905>). *Scientific Reports*. **5** (1): 14272. Bibcode:2015NatSR...514272L (<https://ui.adsabs.harvard.edu/abs/2015NatSR...514272L>). doi:10.1038/srep14272 (<https://doi.org/10.1038/srep14272>). PMC 4585905 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4585905>). PMID 26404901 (<https://pubmed.ncbi.nlm.nih.gov/26404901>).

160. Eduati F, Utharala R, Madhavan D, Neumann UP, Longerich T, Cramer T, et al. (June 2018). "A microfluidics platform for combinatorial drug screening on cancer biopsies" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6015045>). *Nature Communications*. **9** (1): 2434. Bibcode:2018NatCo...9.2434E (<https://ui.adsabs.harvard.edu/abs/2018NatCo...9.2434E>). doi:10.1038/s41467-018-04919-w (<https://doi.org/10.1038/s41467-018-04919-w>). PMC 6015045 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6015045>). PMID 29934552 (<https://pubmed.ncbi.nlm.nih.gov/29934552>).
161. Stevens MM, Maire CL, Chou N, Murakami MA, Knoff DS, Kikuchi Y, et al. (November 2016). "Drug sensitivity of single cancer cells is predicted by changes in mass accumulation rate" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5142231>). *Nature Biotechnology*. **34** (11): 1161–1167. doi:10.1038/nbt.3697 (<https://doi.org/10.1038/nbt.3697>). PMC 5142231 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5142231>). PMID 27723727 (<https://pubmed.ncbi.nlm.nih.gov/27723727>).
162. Sart S, Tomasi RF, Amselem G, Baroud CN (September 2017). "Multiscale cytometry and regulation of 3D cell cultures on a chip" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5589863>). *Nature Communications*. **8** (1): 469. Bibcode:2017NatCo...8..469S (<https://ui.adsabs.harvard.edu/abs/2017NatCo...8..469S>). doi:10.1038/s41467-017-00475-x (<https://doi.org/10.1038/s41467-017-00475-x>). PMC 5589863 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5589863>). PMID 28883466 (<https://pubmed.ncbi.nlm.nih.gov/28883466>).
163. Grosselin K, Durand A, Marsolier J, Poitou A, Marangoni E, Nemati F, et al. (June 2019). "High-throughput single-cell ChIP-seq identifies heterogeneity of chromatin states in breast cancer". *Nature Genetics*. **51** (6): 1060–1066. doi:10.1038/s41588-019-0424-9 (<https://doi.org/10.1038/s41588-019-0424-9>). PMID 31152164 (<https://pubmed.ncbi.nlm.nih.gov/31152164>). S2CID 171094979 (<https://api.semanticscholar.org/CorpusID:171094979>).

Further reading

Review papers

- Yetisen AK, Akram MS, Lowe CR (June 2013). "Paper-based microfluidic point-of-care diagnostic devices". *Lab on a Chip*. **13** (12): 2210–2251. doi:10.1039/C3LC50169H (<https://doi.org/10.1039/C3LC50169H>). PMID 23652632 (<https://pubmed.ncbi.nlm.nih.gov/23652632>). S2CID 17745196 (<https://api.semanticscholar.org/CorpusID:17745196>).
- Whitesides GM (July 2006). "The origins and the future of microfluidics". *Nature*. **442** (7101): 368–373. Bibcode:2006Natur.442..368W (<https://ui.adsabs.harvard.edu/abs/2006Natur.442..368W>). doi:10.1038/nature05058 (<https://doi.org/10.1038/nature05058>). PMID 16871203 (<https://pubmed.ncbi.nlm.nih.gov/16871203>). S2CID 205210989 (<https://api.semanticscholar.org/CorpusID:205210989>).
- Seemann R, Brinkmann M, Pfohl T, Herminghaus S (January 2012). "Droplet based microfluidics". *Reports on Progress in Physics*. **75** (1): 016601. Bibcode:2012RPPPh...75a6601S (<https://ui.adsabs.harvard.edu/abs/2012RPPPh...75a6601S>). doi:10.1088/0034-4885/75/1/016601 (<https://doi.org/10.1088/0034-4885/75/1/016601>). PMID 22790308 (<https://pubmed.ncbi.nlm.nih.gov/22790308>). S2CID 5206697 (<https://api.semanticscholar.org/CorpusID:5206697>).
- Squires TM, Quake SR (2005). "Microfluidics: Fluid physics at the nanoliter scale" (<http://authors.library.caltech.edu/1310/1/SQUrmp05.pdf>) (PDF). *Reviews of Modern Physics*. **77** (3): 977–1026. Bibcode:2005RvMP...77..977S (<https://ui.adsabs.harvard.edu/abs/2005RvMP...77..977S>). doi:10.1103/RevModPhys.77.977 (<https://doi.org/10.1103/RevModPhys.77.977>).

- Yetisen AK, Volpatti LR (July 2014). "Patent protection and licensing in microfluidics". *Lab on a Chip*. **14** (13): 2217–2225. doi:10.1039/C4LC00399C (<https://doi.org/10.1039%2FC4LC00399C>). PMID 24825780 (<https://pubmed.ncbi.nlm.nih.gov/24825780>). S2CID 8669721 (<https://api.semanticscholar.org/CorpusID:8669721>).
- Chen K (2011). "Microfluidics and the future of drug research" (<https://web.archive.org/web/20120331110240/http://juls.library.utoronto.ca/index.php/juls/article/view/14551/12241>). *Journal of Undergraduate Life Sciences*. **5** (1): 66–69. Archived from the original (<http://juls.library.utoronto.ca/index.php/juls/article/view/14551/12241>) on 2012-03-31. Retrieved 2011-08-30.
- Angell JB, Terry SC, Barth PW (April 1983). "Silicon Micromechanical Devices". *Scientific American*. **248** (4): 44–55. Bibcode:1983SciAm.248d..44A (<https://ui.adsabs.harvard.edu/abs/1983SciAm.248d..44A>). doi:10.1038/scientificamerican0483-44 (<https://doi.org/10.1038%2Fscientificamerican0483-44>).
- Carugo D, Bottaro E, Owen J, Stride E, Nastruzzi C (May 2016). "Liposome production by microfluidics: potential and limiting factors" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4872163>). *Scientific Reports*. **6**: 25876. Bibcode:2016NatSR...625876C (<https://ui.adsabs.harvard.edu/abs/2016NatSR...625876C>). doi:10.1038/srep25876 (<https://doi.org/10.1038%2Fsrp25876>). PMC 4872163 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4872163>). PMID 27194474 (<https://pubmed.ncbi.nlm.nih.gov/27194474>).
- Chossat JB, Park YL, Wood RJ, Duchaine V (September 2013). "A Soft Strain Sensor Based on Ionic and Metal Liquids". *IEEE Sensors Journal*. **13** (9): 3405–3414. Bibcode:2013ISenJ..13.3405C (<https://ui.adsabs.harvard.edu/abs/2013ISenJ..13.3405C>). CiteSeerX 10.1.1.640.4976 (<https://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.640.4976>). doi:10.1109/JSEN.2013.2263797 (<https://doi.org/10.1109%2FJSEN.2013.2263797>). S2CID 14492585 (<https://api.semanticscholar.org/CorpusID:14492585>).
- Tseng TM, Li M, Freitas DN, Mongersun A, Araci IE, Ho TY, Schlichtmann U (2018). *Columba S: a scalable co-layout design automation tool for microfluidic large-scale integration* (https://edawww.regent.e-technik.tu-muenchen.de/public/upload/201807122255_DAC18_ColumbaS_Tsun-Ming.pdf) (PDF). Proceedings of the 55th Annual Design Automation Conference. p. 163.

Books

- Bruus H (2008). *Theoretical Microfluidics*. Oxford University Press. ISBN 978-0199235094.
- Folch, Albert. *Hidden in Plain Sight: The History, Science, and Engineering of Microfluidic Technology* (MIT Press, 2022) online review (<http://www.h-net.org/reviews/showrev.php?id=57939>)
- Herold KE, Rasooly A (2009). *Lab-on-a-Chip Technology: Fabrication and Microfluidics*. Caister Academic Press. ISBN 978-1-904455-46-2.
- Kelly R, ed. (2012). *Advances in Microfluidics*. Richland, Washington, USA: Pacific Northwest National Laboratory. ISBN 978-953-510-106-2.
- Jenkins G, Mansfield CD (2012). *Microfluidic Diagnostics*. Humana Press. ISBN 978-1-62703-133-2.
- Li X, Zhou Y, eds. (2013). *Microfluidic devices for biomedical applications*. Woodhead Publishing. ISBN 978-0-85709-697-5.
- Tabeling P (2006). *Introduction to Microfluidics* (<https://archive.org/details/introductiontomi000tabe>). Oxford UP. ISBN 978-0-19-856864-3.

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